

DVLD

DRIVING & VEHICLE LICENSE DEPARTMENT SYSTEM

DVLD – DVLD – Driving & Vehicle License Department System

[Document subtitle]

Abd Alrahman Saed

1/1/25

Desktop Application

Introduction:

The DVLD Management System is a robust desktop application engineered to **streamline and automate the core processes** of a governmental Driving and Vehicle License Department. Its primary objective is to digitalize all critical operations, ensuring a scalable, maintainable, and efficient solution for managing citizen records, license applications, driving tests, and driver history.

System Modules / Structure

The application is structured into a logical set of modules, reflecting the real-world operations of a licensing authority.

- **People Management:** Centralized module for creating, viewing, updating, and deleting individual citizen records, as well as providing a detailed "Person Information Card."
- **Licensing Services & Applications:** The main hub for all license-related workflows.
  - ✓ New Local Driving License
  - ✓ International Driving License
  - ✓ Renew Local License
  - ✓ Replace Lost or Damaged License
  - ✓ Release Detained License
  - ✓ Retake Test
- **Drivers Management:** Provides a comprehensive list and history of all registered drivers.

- **Tests Management:** Handles the scheduling and recording of driving tests, including the management of various Test Types (e.g., Vision Test, Written Test, Practical Test).
- **Users and Account Settings:** Manages system user accounts, user permissions, and provides functionality for users to view their profile and change their password.

User Roles

While specific role names can be defined, the system architecture facilitates separation of duties through implied roles:

- **Admin/User Manager:** Manages system users and application configuration (e.g., Test Types).
- **Licensing Officer/Application Manager:** Handles the processing and management of all driving license applications and services.
- **Driver/People Viewer:** Users who can access and view driver or person information records, possibly with read-only privileges.
- **Tester:** Responsible for scheduling and logging the results of driving tests.

Technology Stack

The DVLD Management System is a high-quality Windows desktop application adhering to modern software engineering principles:

- Programming Language: **C#**
- Framework: **.NET Framework** (targeting v4.8)
- Architecture: Multi-layered **(3-Tier) Architecture**, ensuring a clean separation of concerns:
  - ✓ **Presentation Layer:** Handles the **user interface** (UI) and user interactions.
  - ✓ **Business Layer:** Contains the **core business logic** and rules.
  - ✓ **Data Access Layer:** Manages all **communication with the database** (SQL Server is the implied database).

Project Scope

The project's scope covers the end-to-end management of driving license services within the department. This includes:

- Comprehensive **record-keeping for all people** and drivers.
- **Management of the entire lifecycle for all license applications** (initial, renewal, replacement).
- **Handling of detained licenses**, including detaining and release processes.
- **Configuration of application and test types** and their associated fees.

Navigation Overview (High-Level Map)

The main application screen provides the user with a menu-driven navigation system that groups functionalities logically:

Main Menu Item	Primary Functionalities
Applications	Driving Licenses Services: ✓ New Local/International ✓ Renew License ✓ Replace License ✓ Release Detained Licenses ✓ Retake Test ✓ Manage Applications ✓ Manage Detained Licenses ✓ Manage Test Types
People	Access and manage the list of all individual records.
Drivers	Access and manage the list of all registered drivers and their license history.
Users	Manage system user accounts and permissions.
Account Settings	Current User Profile, Change Password, Sign Out.

Screens Documentation:

1- Login Module

Document Version: **2.0**

Last Updated: **12 Dec 2025**

Author: Abd Alrahman Saed

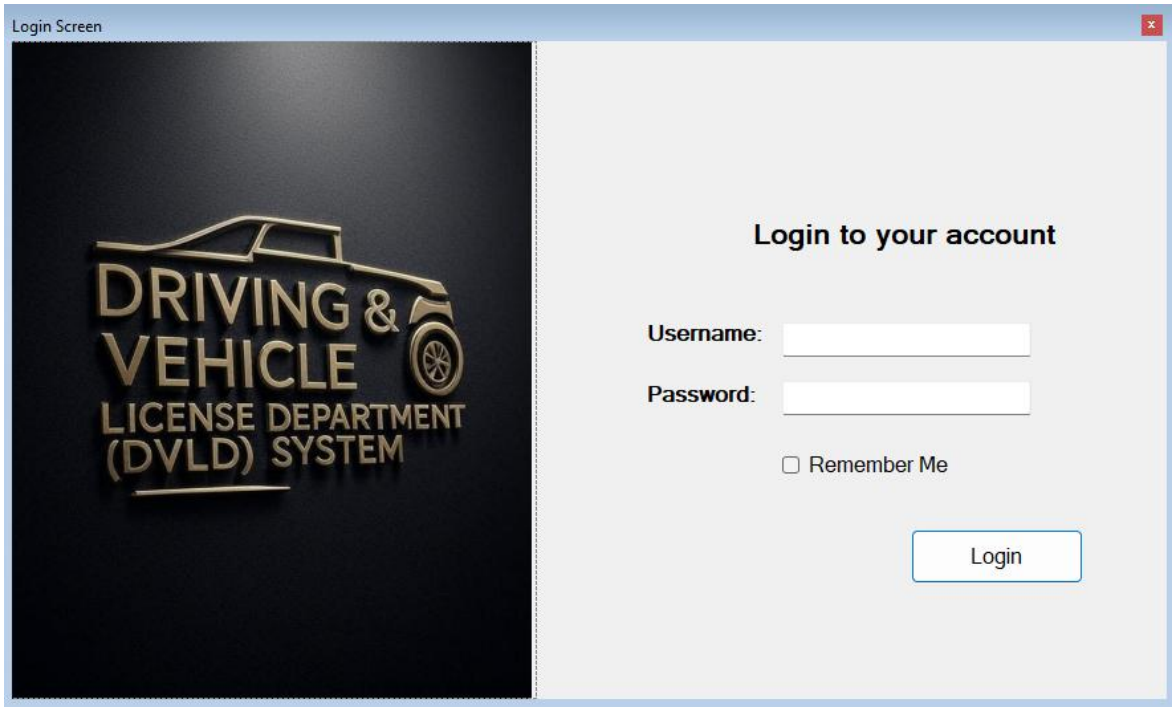
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Login Module**

Screen ID: **UI-Log-001**

Screenshot:



Screen Purpose:

Authenticate system users by validating their credentials (**username and password**), **checking account status**, and granting access to the main system. Supports "**Remember Me**" functionality and protects the system from unauthorized access.

Functions:

- ✓ Validate Username
- ✓ User Authentication
- ✓ Remember Me
- ✓ Error Handling
- ✓ Navigation to Main Screen

Validations:

- ✓ Username required
- ✓ Password required
- ✓ User must exist in the system
- ✓ Account must be active
- ✓ Show inline error messages using ErrorProvider

Navigation:

System Start → Login Screen → Main Form

If login fails → Stay on Login Screen

If inactive account → Show error and stay on Login Screen

Error Messages:

Error Code	Message	Trigger
ERR-001	Please enter Username	Empty Username field
ERR-002	Please enter Password	Empty Password field
ERR-003	Invalid Username or Password	Credentials not found
ERR-004	Your account is not Active, Contact Admin.	Logged user's IsActive = false
ERR-005	Some fields are required.	Validation failed on login button

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Expected Outputs:

- ✓ **Successful Login:** Remember Me saved (if checked)→ Open Main Screen→ Hide Login screen
- ✓ **Failed Login:** Inline validation error OR Message Box: Invalid credentials → No navigation occurs
- ✓ **Remember Me:** On load: Username & Password pre-filled, Check Box state restored

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Changes Log:

Version	Date	Description of Change	Author
1.0	03 Dec 2025	Initial creation of Login Screen documentation.	Abd Alrahman Saed
2.0	12 Dec 2025	Remove UI Component Tabel (Useless)	Abd Alrahman Saed

2. Main Menu

Document Version: **2.0**

Last Updated: **12 Dec 2025**

Author: Abd Alrahman Saed

Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

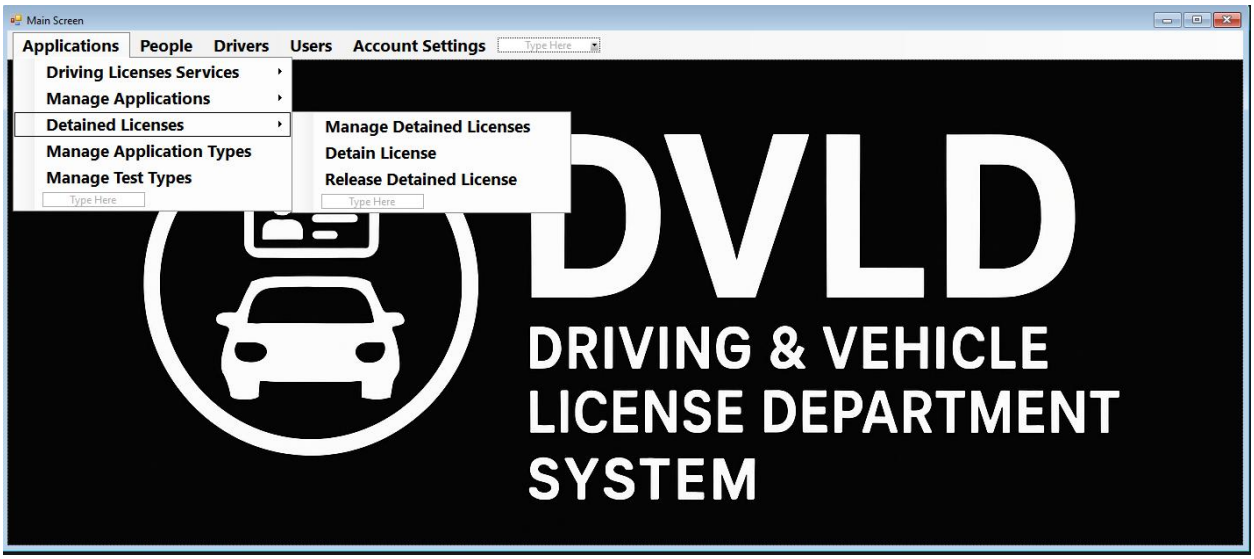
Module Name: **Main Module**

Screen ID: **UI-Main-002**

Screenshot:









Screen Purpose:

Provides centralized access to all system modules including People, Users, Applications, Licenses, Tests, Detained Licenses, and Settings.

Functions:

- ✓ Open People List
- ✓ Open Users List
- ✓ Show Current User Information
- ✓ Change Password
- ✓ Logout
- ✓ Manage Applications
- ✓ Manage Test Types
- ✓ Add Local Driving License Application
- ✓ List of Local Driving License Applications
- ✓ List of Drivers
- ✓ List International License Applications
- ✓ Renew Local Driving License
- ✓ Replace Lost/Damaged License
- ✓ Detain a License
- ✓ Release Detained License
- ✓ List of Detained Licenses
- ✓ Issue International License

Validations:

(No direct input fields on this screen — no validations required.)

Navigation: Login → Main Menu

Main Menu → All System Modules (People, Users, Applications, Licenses, Drivers, Tests)

Error Messages: No Error Messages

Expected Outputs:

- ✓ Opens selected module screen as a modal dialog
- ✓ Logs out the user and returns to Login form
- ✓ Displays user-specific options (Current User Info, Change Password)

Changes Log:

Version	Date	Description of Change	Author
1.0	04 Dec 2025	Initial creation of Main Menu screen documentation.	Abd Alrahman Saed
2.0	12 Dec 2025	Remove UI Component Tabel (Useless)	Abd Alrahman Saed

3. Add / Edit Person Information Screen

Document Version: **2.0**

Last Updated: **12 Dec 2025**

Author: Abd Alrahman Saed

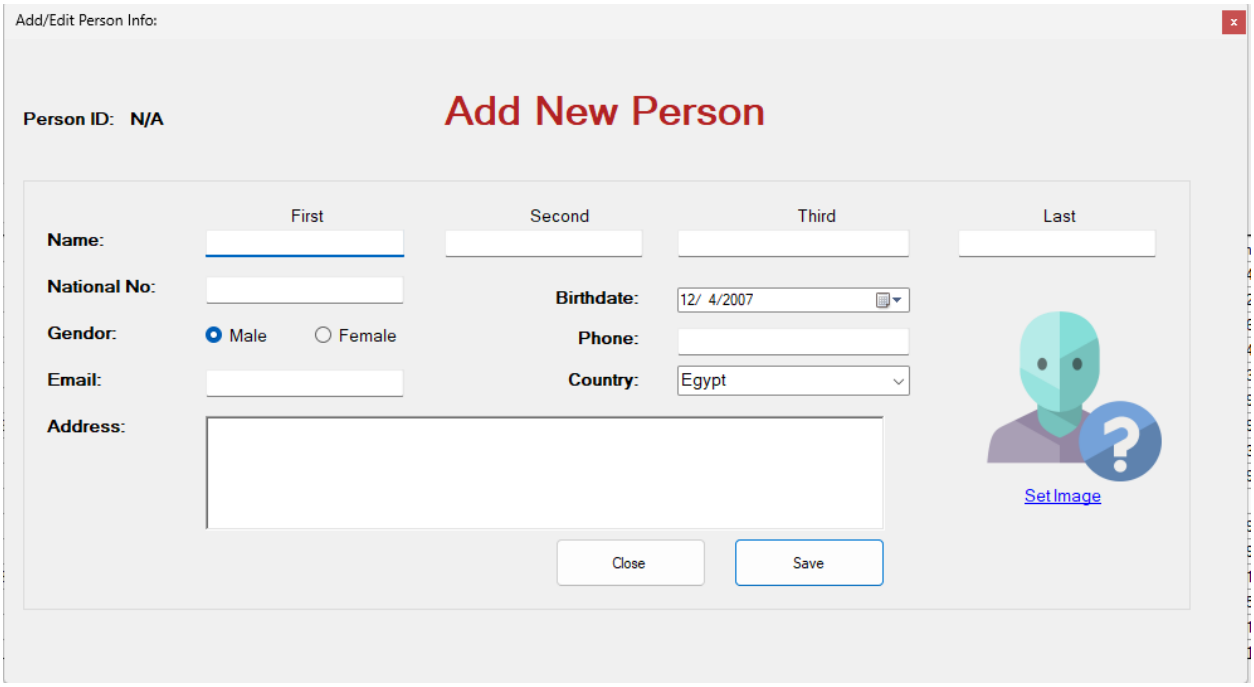
Reviewer: Dr. Heba Rashed

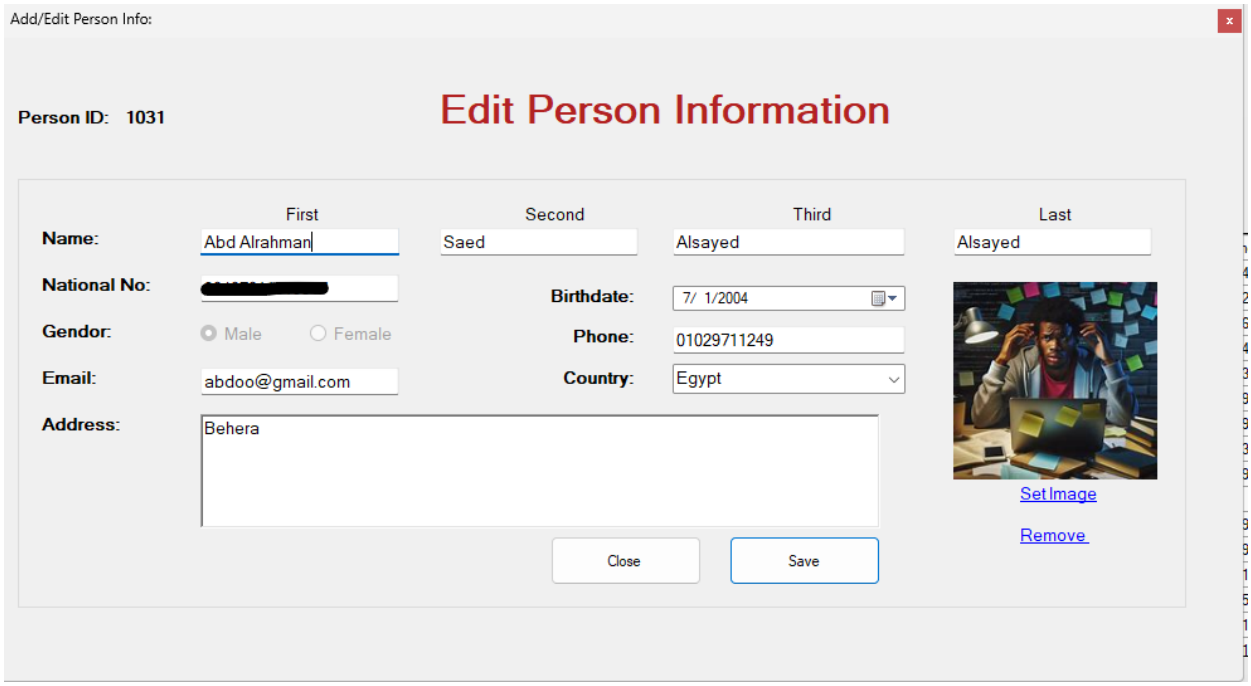
System Name: DVLD – Driving & Vehicle License Department System

Module Name: **People Module**

Screen ID: **UI-Pep-003**

Screenshot:





Screen Purpose:

This screen is used to **add or edit a person's information** in the DVLD system.
It collects complete personal data including name, contact details, address, and personal identifiers (National Number, Date of Birth, Gender, etc.).

Functions:

- ✓ Add new person
- ✓ Edit existing person
- ✓ Validate all input fields
- ✓ Check if National Number already exists
- ✓ Check if Email already used
- ✓ Auto-fill fields when editing
- ✓ Save data to database

Validations:

- ✓ First name required
- ✓ Last name required
- ✓ National Number required
- ✓ National Number must be unique
- ✓ Phone must be numeric
- ✓ Phone must be 11 digits
- ✓ Email must be valid format
- ✓ Email must not be duplicated
- ✓ Date of Birth must indicate age ≥ 18
- ✓ Gender required
- ✓ Address required

Navigation:

- ✓ Main Menu → People → Add Person
- ✓ Main Menu → People → Edit Person

Error Messages:

Error Code	Message	Trigger
ERR-006	First name cannot be empty.	First name missing
ERR-007	Last name cannot be empty.	Last name missing
ERR-008	National Number cannot be empty.	National No missing
ERR-009	National Number must be numeric.	Contains non-numeric chars
ERR-010	National Number already exists.	Duplicate National No
ERR-011	Phone number must be numeric.	Phone contains letters
ERR-012	Phone number must be 11 digits.	Invalid length

Error Code	Message	Trigger
ERR-013	Email format is invalid.	Invalid email format
ERR-014	Email already exists.	Duplicate email
ERR-015	Please select gender.	Gender not chosen
ERR-016	Address cannot be empty.	Address missing
ERR-017	Person must be 18 years old or older.	Age < 18
ERR-018	Please fill all required fields.	General missing data

---

Expected Outputs:

- ✓ **Add New Person:** Data saved → success message → form closes → people list refresh
- ✓ **Update Person Info:** Updated successfully → form closes → people list refresh
- ✓ **Invalid input:** Show appropriate error message
- ✓ **Duplicate identifiers:** Block save and display error
- ✓ **Cancel operation:** Close form without saving

Changes Log:

Version	Date	Description of Change	Author
1.0	04 Dec 2025	Initial creation of Add/Edit Person screen documentation	Abd Alrahman Saed
2.0	12 Dec 2025	Remove UI Component Tabel (Useless)	Abd Alrahman Saed

4. Show Person Info

Document Version: **1.0**

Last Updated: **13 Dec 2025**

Author: Abd Alrahman Saed

Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **People Module**

Screen ID: **UI-PER-004**

Screenshot:



Screen Purpose:

Display complete person information in read-only mode. Reuse ctrlPersonCard for consistency. Used as a reference screen across the system.

Functions:

- ✓ Show Person Information

Validations:

- ✓ Internal Validate Is there a Person Exists with the ID Passed or Not.

Navigation:

Main Menu → People → Show Person Info

Any screen → View Person → Opens this form

Error Messages:

This screen is **read-only** — all errors are system-related.

Error Code	Message	Trigger
ERR-023	Person not found.	Invalid PersonID / NationalNo
ERR-024	Failed to load person information.	Database error
ERR-026	Missing or corrupted person data.	Incomplete data

Expected Outputs:

- Person information displayed correctly

Changes Log:

Version	Date	Description of Change	Author
1.0	13 Dec 2025	Initial Creation of Show Person Info screen	Abd Alrahman Saed

5. Find Person

Document Version: **1.0**

Last Updated: **13 Dec 2025**

Author: Abd Alrahman Saed

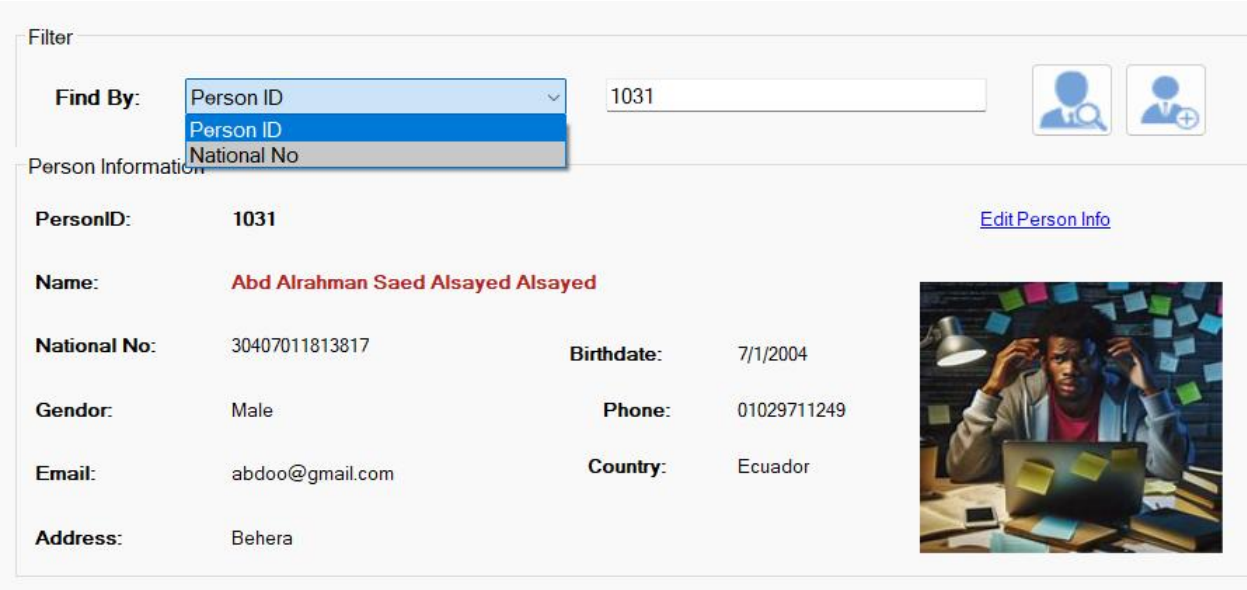
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **People Module**

Screen ID: **UI-PER-005**

Screenshot:



Screen Purpose:

- ✓ Search for a person using **Person ID** or **National Number**.
- ✓ Display person details using embedded ctrlPersonCard
- ✓ Allow selecting a person and notifying parent form
- ✓ Allow adding a new person (Optionally)

Functions:

- ✓ Find
- ✓ Display Information
- ✓ Add New Person

Validations:

- ✓ Name required
- ✓ Fees numeric
- ✓ Driver must exist

Navigation:

- ✓ Local Driving License Application
- ✓ International License Application
- ✓ Any form requiring person selection

Error Messages:

Error Code	Message	Trigger
ERR-023	Person not found.	Invalid Person ID / National No
ERR-024	Failed to load person information.	Database error

Expected Outputs:

- ✓ Person data loaded correctly
- ✓ Event fired with selected PersonID

Changes Log:

Version	Date	Description of Change	Author
1.0	13 Dec 2025	Initial creation of Find Person screen.	Abd Alrahman Saed

6. List & Manage People

Document Version: **1.0**

Last Updated: **13 Dec 2025**

Author: Abd Alrahman Saed

Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **People Module**

Screen ID: **UI-PER-006**

Screenshot:





Screen Purpose:

- ✓ Display a list of all registered people in the DVLD system.
- ✓ Allow users to search, filter, add, edit, delete, and view detailed person information.
- ✓ Act as the central management screen for all person-related operations.

Functions:

- ✓ Load all people records on form load
- ✓ Display selected columns only (clean view)
- ✓ Filter people by:
    - Person ID
    - National No
    - Phone
    - First / Second / Third / Last Name
    - Country
    - Gender
- ✓ Add new person
- ✓ Edit existing person
- ✓ Delete selected person
- ✓ Show detailed person information

Validations: No Direct Validation from user

Navigation: Main Menu → People → Manage People

Expected Outputs:

- ✓ People list loaded successfully
- ✓ Accurate filtering and searching
- ✓ Successful CRUD operations
- ✓ Correct record count displayed

Changes Log:

Version	Date	Description of Change	Author
1.0	13 Dec 2025	Initial creation of List & Manage People screen.	Abd Alrahman Saed

7. List Application Types

Document Version: **1.0**

Last Updated: **13 Dec 2025**

Author: Abd Alrahman Saed

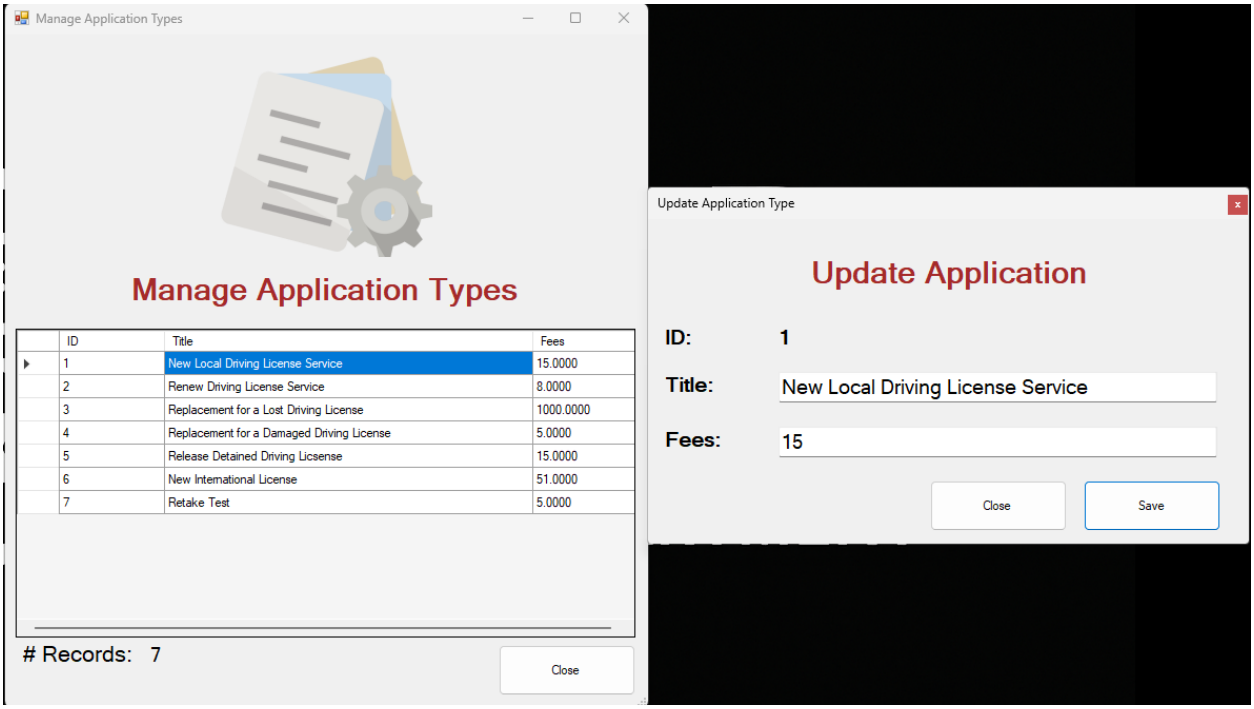
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Application Types Module**

Screen ID: **UI-APP_TYP-007**

Screenshot:



Screen Purpose:

- ✓ Display all application types available in the DVLD system.
- ✓ Allow **Admin users only** to update application type details (Title & Fees).
- ✓ Provide a read-only view for non-admin users.

Functions:

- ✓ Load all application types on form load
- ✓ Restrict update operation to **Admin users only**
- ✓ Open update form for selected application type

Validations:

- ✓ Name required
- ✓ Fees numeric
- ✓ Driver must exist

Navigation: Main Menu → Applications → Manage Application Types

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Error Messages:

Error Code	Message	Trigger
ERR-025	Failed to load application types.	Database error while loading data
ERR-026	Access denied. Admin privileges required.	Non-admin user attempts update

---

Expected Outputs:

- ✓ Application types loaded successfully
- ✓ Correct number of records displayed.
- ✓ Admin users can update applications

Changes Log:

Version	Date	Description of Change	Author
1.0	13 Dec 2025	Initial documentation for List Application Types screen	Abd Alrahman Saed

8. Add / Edit Local Driving License Application

Document Version: **2.0**

Last Updated: **12 Dec 2025**

Author: Abd Alrahman Saed

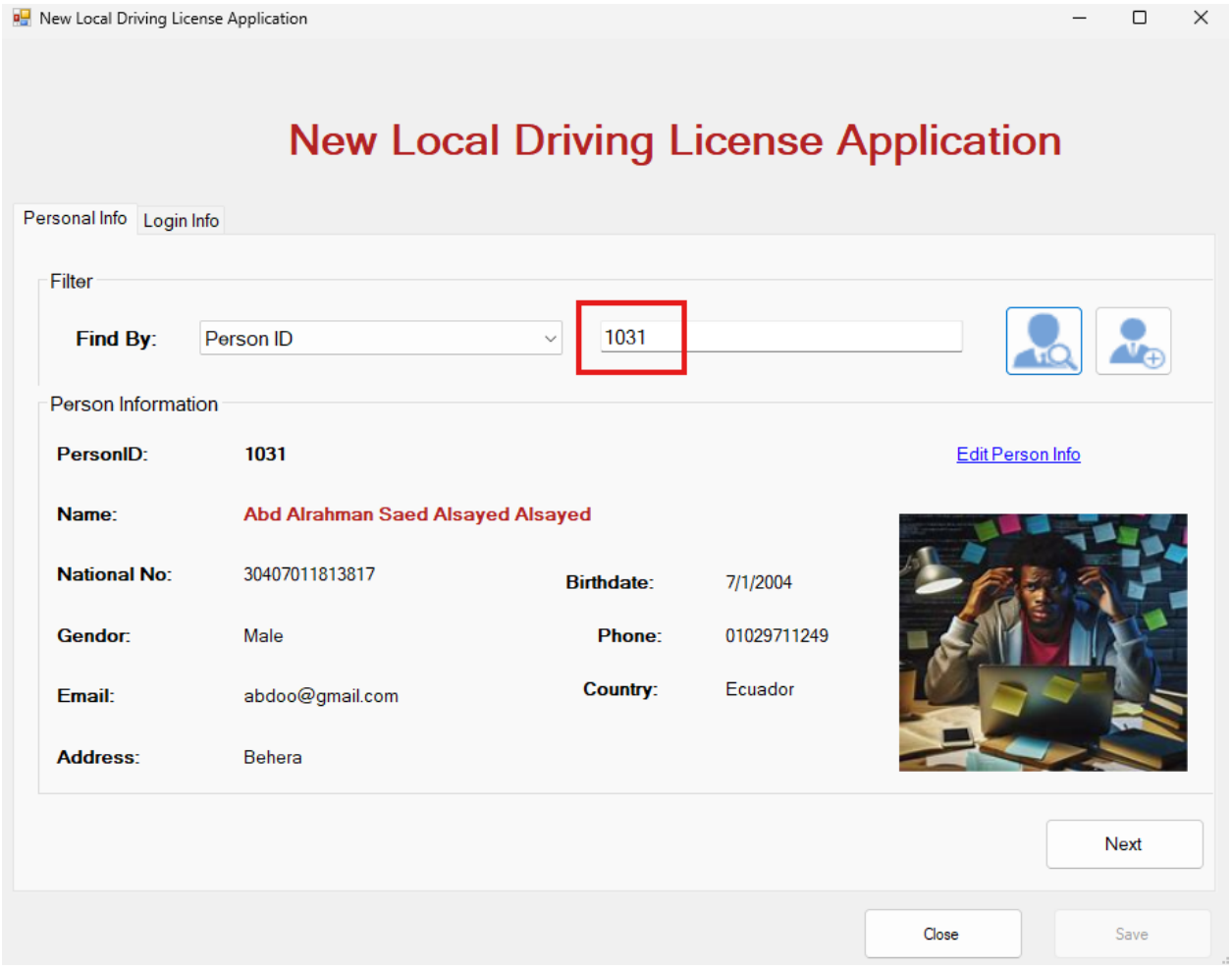
Reviewer: Dr. Heba Rashed

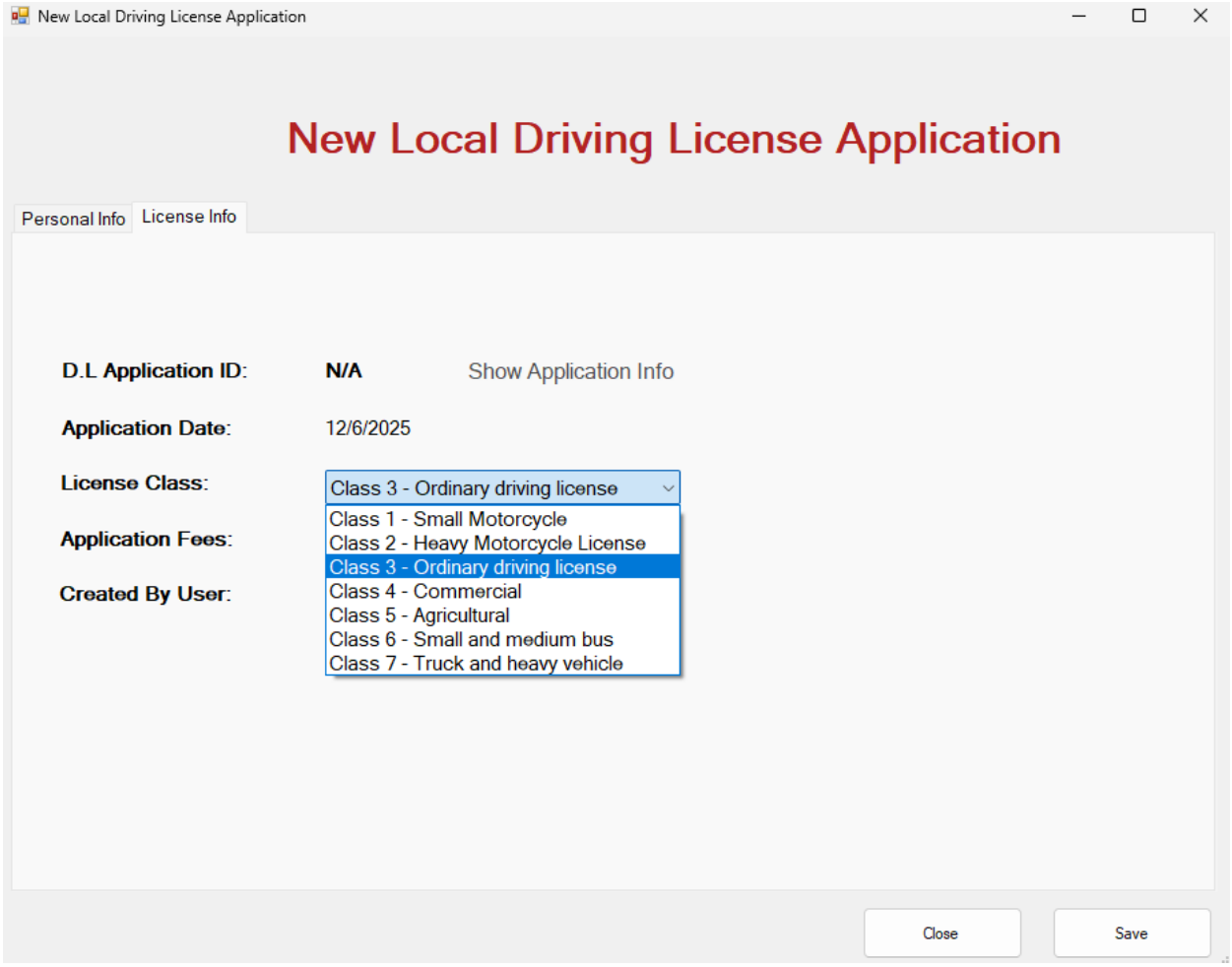
System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Local Driving License Application Module**

Screen ID: **UI-LDL-008**

Screenshot:





Screen Purpose:

- ✓ Add a new Local Driving License Application.
- ✓ Edit an existing Local Driving License Application.
- ✓ Validate eligibility (age, existing license, active application).
- ✓ Manage license class, applicant information, and application fees.

Functions:

- ✓ Add new Local Driving License Application
- ✓ Edit existing Application
- ✓ Load existing person data
- ✓ Save application
- ✓ Auto-fill fees & creator info

Validations:

- ✓ Person must be selected
- ✓ No active application for the same class
- ✓ No active license for selected class
- ✓ Age meets minimum requirement
- ✓ License class must be chosen

Navigation:

Main Menu → Applications → Driving License Services → New Driving License → Local License

---

Error Messages:

Error Code	Message	Trigger
ERR-014	Please select a valid person for the application.	No person selected when clicking Next
ERR-015	This person already has an active application for the selected license class.	Active application exists
ERR-016	This person already has an active license for the selected license class.	Existing license found

Error Code	Message	Trigger
ERR-017	The person does not meet the minimum age requirement for this license class.	Age < Minimum allowed
ERR-018	Error loading application information.	Application not found on Edit
ERR-019	Error saving application information.	Database save failed

---

Expected Outputs:

- **Add New Application:**
    - Application saved → ID generated → header switches to Edit Mode.
    - Fees auto loaded.
    - Person selection enabled.
- **Edit Application:**
    - Loads previous data (person + license class + fees + created by).
- **Prevents saving if:**
    - Active application exists
    - Person has active license
    - Age requirement not met
    - Person not selected
- **UI:**
    - Next → moves to Login Info tab
    - Save → saves & refreshes screen

Changes Log:

Version	Date	Description of Change	Author
1.0	05 Dec 2025	Initial documentation for Add/Edit Local Driving License Application screen.	Abd Alrahman Saed
2.0	12 Dec 2025	Remove UI Component Tabel (Useless)	Abd Alrahman Saed

9. Show Local Driving License Application Info

Document Version: **2.0**

Last Updated: **12 Dec 2025**

Author: Abd Alrahman Saed

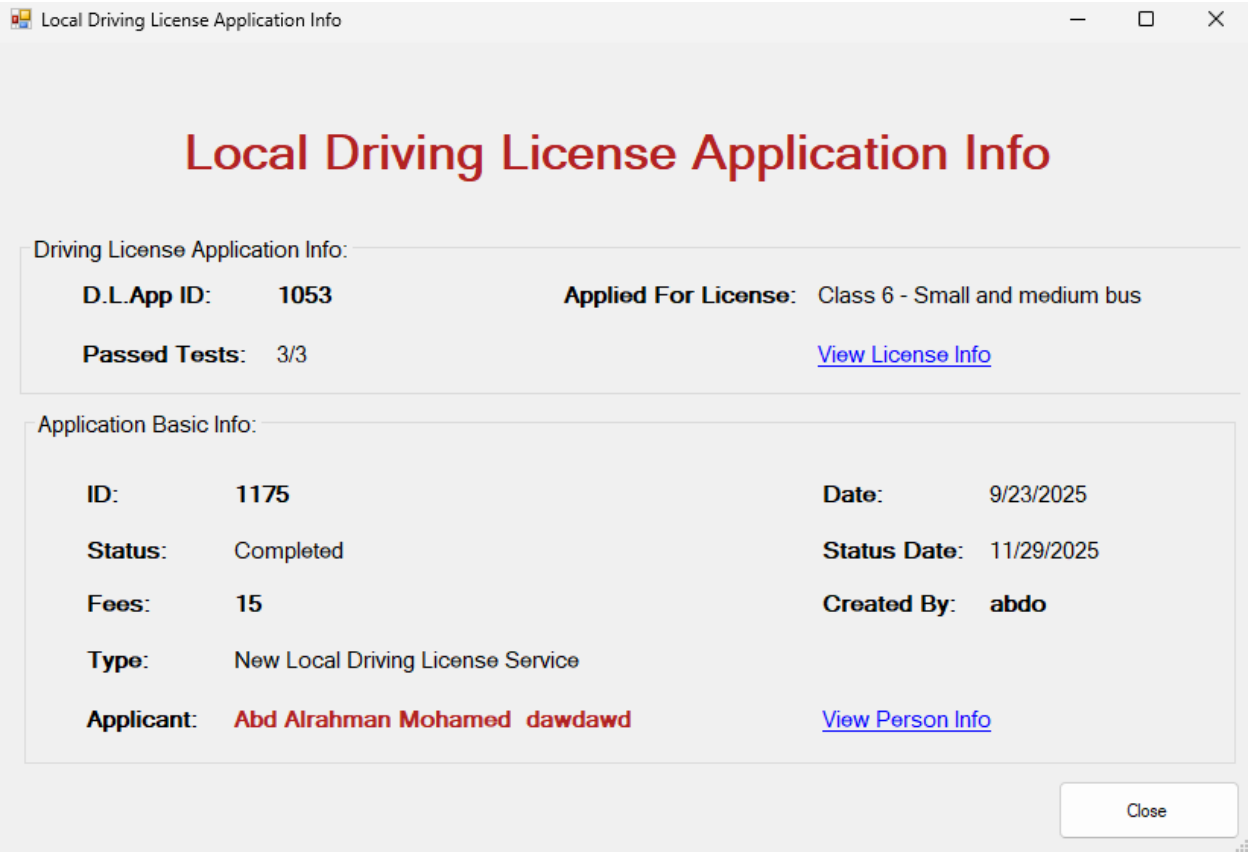
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Local Driving License Application Module**

Screen ID: **UI-LDL-009**

Screenshot:



Screen Purpose:

Display all details of a selected **Local Driving License Application** in read-only mode. Provide fast access to application information: (personal data, class info, fees, application status, and creation metadata...etc).

Used In:

- ✓ Application listing screen
- ✓ Passing test result forms
- ✓ License issuing steps
- ✓ Admin review

Functions:

- ✓ Load LDL Application info by ID at form initialization
- ✓ Display all related data including:
  - Application details
  - License class
  - Applicant personal information
  - Application dates
  - Fees & status

Validations:

- ✓ Name required
- ✓ Fees numeric
- ✓ Driver must exist

Navigation:

- ✓ Main Menu → Applications → Manage Applications → Local Driving License Applications → Right Click on Record → Show Application Details

---

Error Messages:

(No direct user validation errors, All System/Loading Only)

Error Code	Message	Trigger
ERR-020	Application not found.	Invalid LDLApplicationID passed to form
ERR-021	Failed to load application information.	Database error while loading

---

Expected Outputs:

- ✓ Application information loaded successfully
- ✓ User sees all data in read-only mode
- ✓ No editing allowed
- ✓ Close button shuts the form correctly

Changes Log:

Version	Date	Description of Change	Author
1.0	06 Dec 2025	Initial documentation for Show Local Driving License Application Info screen.	Abd Alrahman Saed
2.0	12 Dec 2025	Remove UI Component Tabel (Useless)	Abd Alrahman Saed

10. List / Manage Local Driving License Applications

Document Version: **1.0**

Last Updated: **13 Dec 2025**

Author: Abd Alrahman Saed

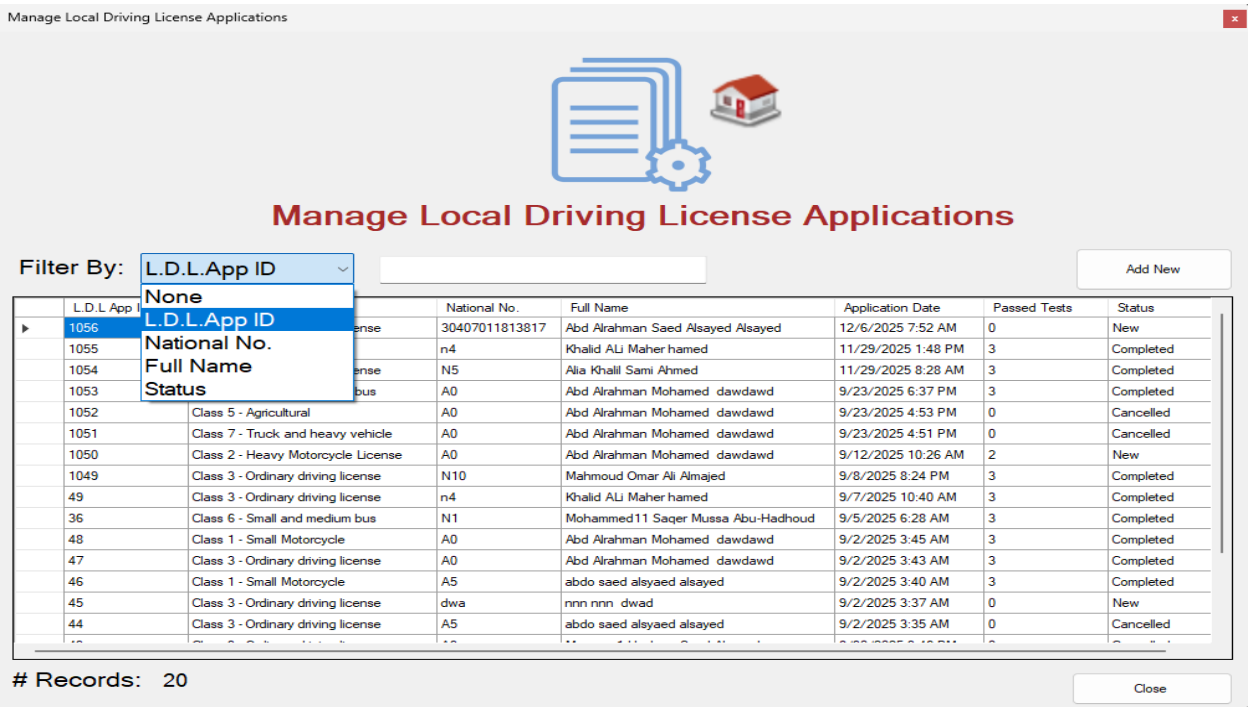
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Local Driving License Application Module**

Screen ID: **UI- LDL -010**

Screenshot:



Screen Purpose:

- ✓ Display all Local Driving License Applications in the system.
- ✓ Provide full control over the application lifecycle including:
  - Searching and filtering
  - Adding new applications
  - Editing existing applications
  - Canceling applications
  - Scheduling required tests
  - Issuing driving licenses
  - Viewing application and license details
- ✓ This screen acts as the **central management hub** for Local Driving License Applications.

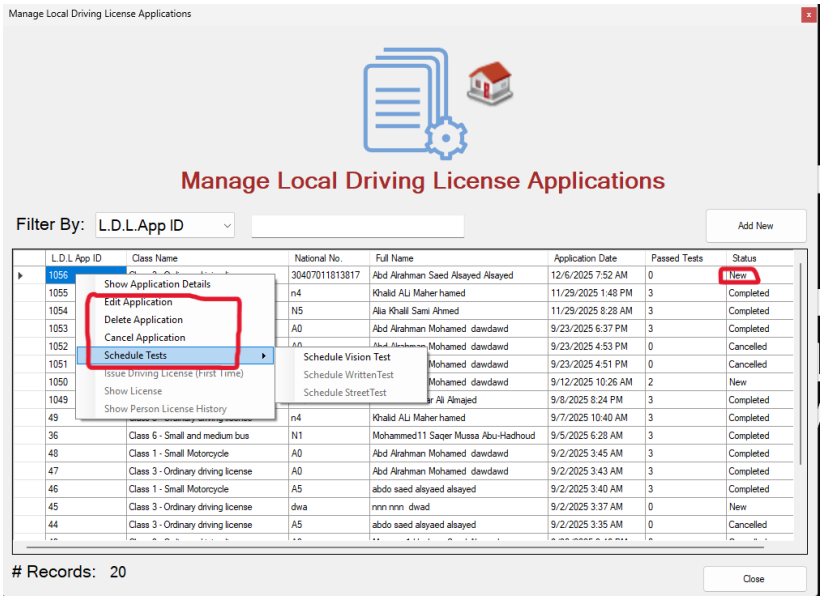
Functions:

- ✓ Load all Local Driving License Applications on form load
- ✓ Filter applications by:
  - Application ID
  - National Number
  - Full Name
  - Status
- ✓ Add new Local Driving License Application
- ✓ Edit application when allowed
- ✓ Cancel applications
- ✓ Enforce correct test scheduling order
- ✓ Issue driving license after all tests are passed
- ✓ Display license and license history
- ✓ Dynamically enable/disable actions based on application state

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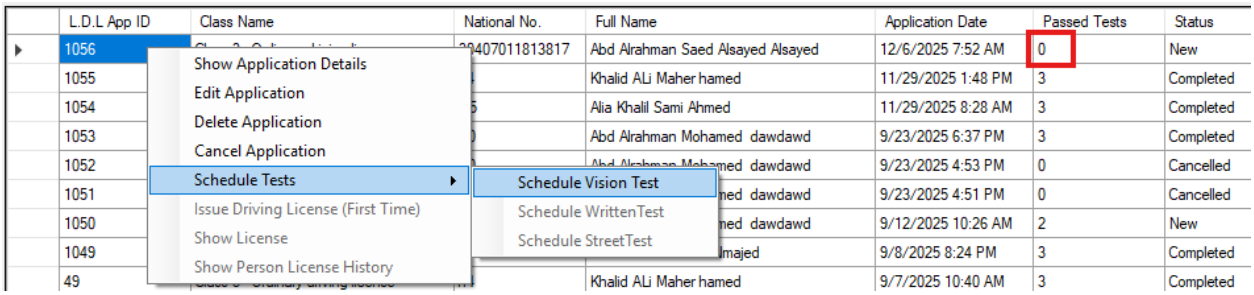
Business Rules:

- ✓ Only applications with status New can be edited or canceled

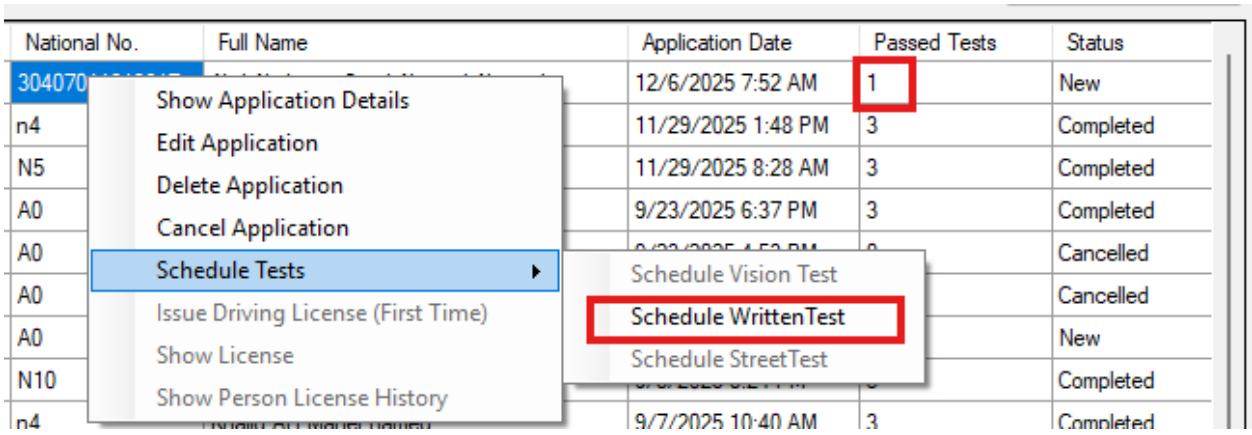


- ✓ **Tests must be taken in strict order:**

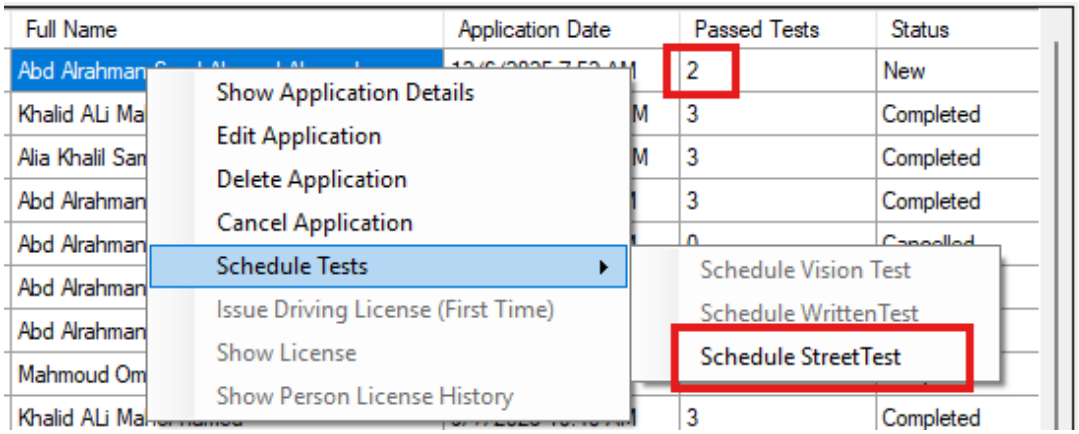
1. **Vision Test**



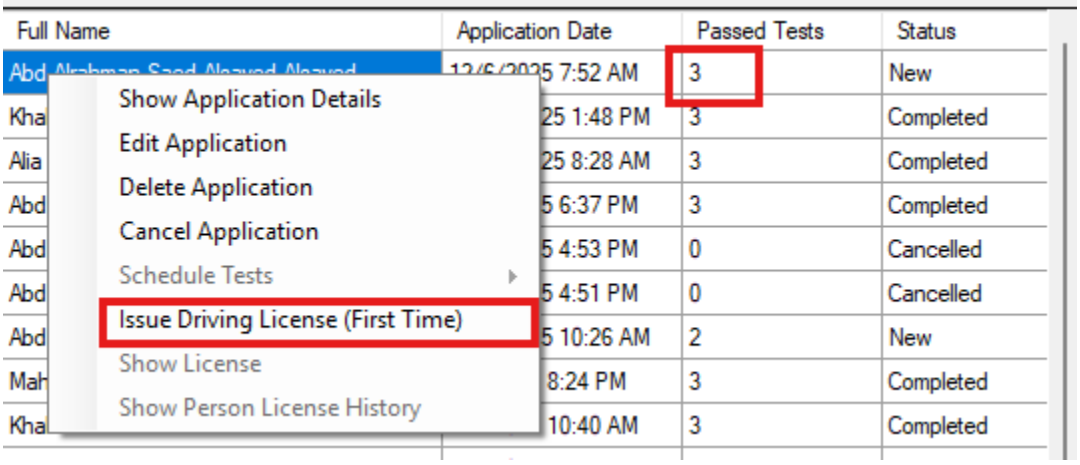
2. **Written Test**



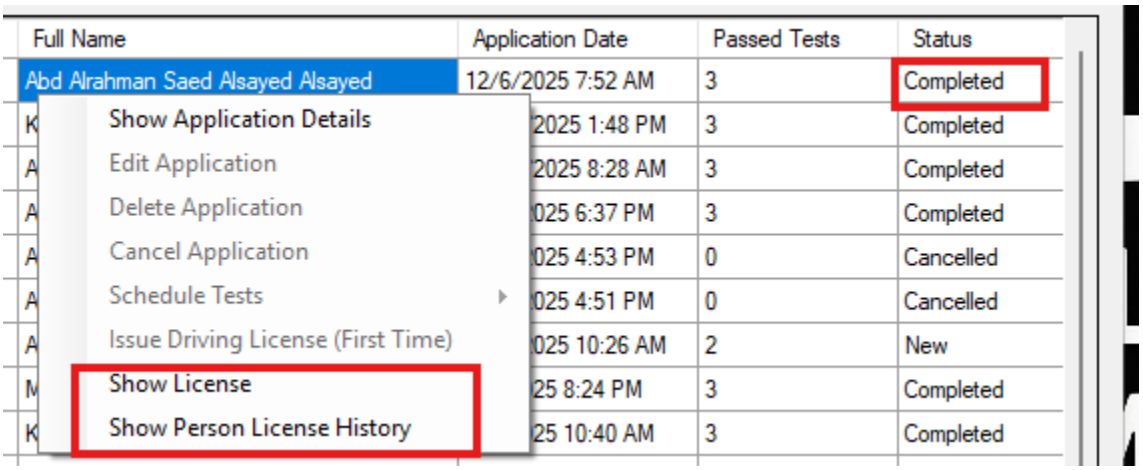
3. **Street Test**



✓ **Driving license can only be issued after all tests are passed**



- ✓ Once a license is issued:



- ✓ Test scheduling becomes disabled
- ✓ License viewing options become enable

Navigation: Main Menu → Applications → Manage Applications → Local Driving License

---

Expected Outputs:

- ✓ Applications list loaded successfully
- ✓ Accurate filtering and searching
- ✓ Correct action availability per application state
- ✓ Successful navigation between related screens
- ✓ Consistent and reliable application management

Changes Log:

Version	Date	Description of Change	Author
1.0	02 Dec 2025	Initial documentation for List Local Driving License Applications screen.	Abd Alrahman Saed

11. Issue International Driving License

Document Version: **1.0**

Last Updated: **02 Dec 2025**

Author: Abd Alrahman Saed

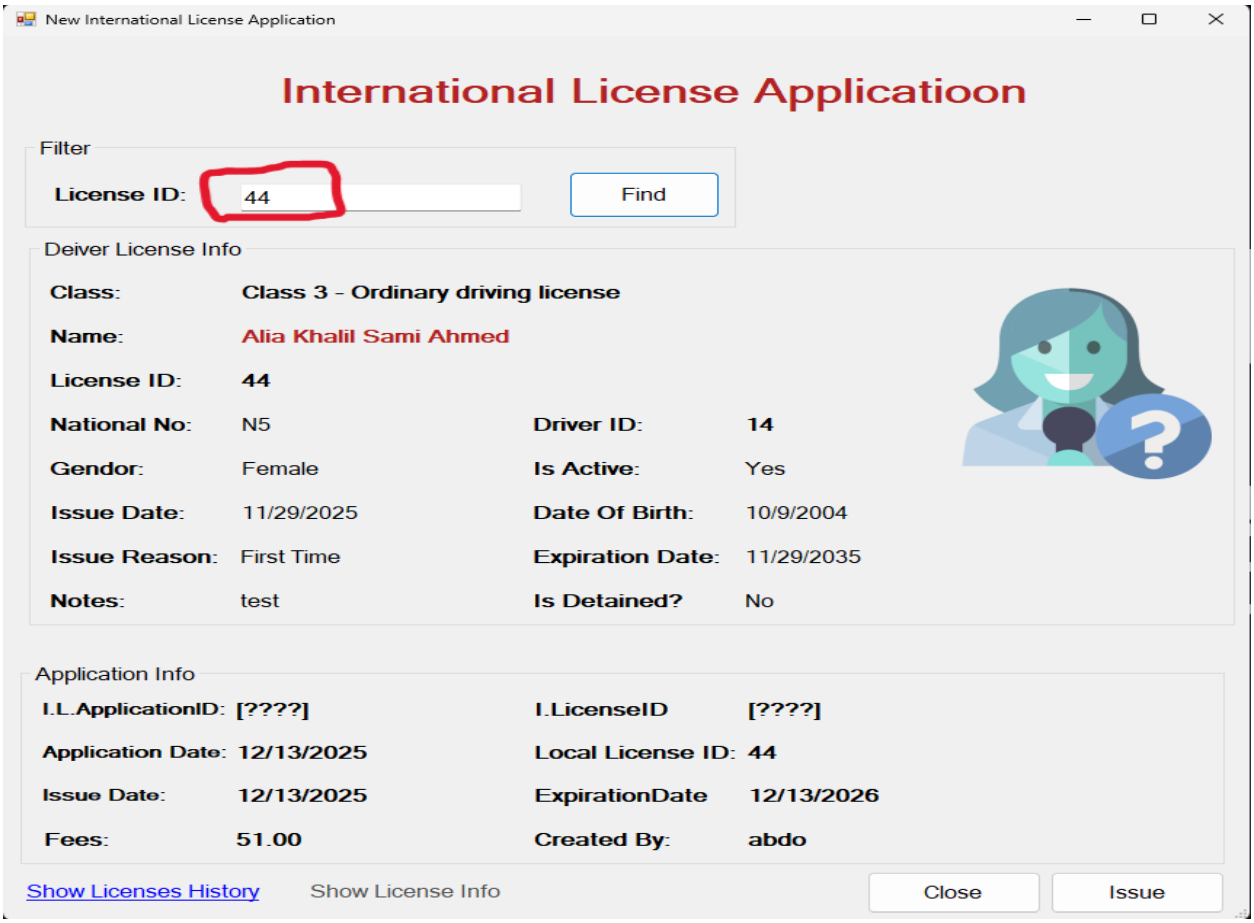
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **International Driving License Module**

Screen ID: **UI-IDL-011**

Screenshot:



Screen Purpose:

- ✓ Issue a new **International Driving License** based on an existing **valid class 3 Local Driving License**.
- ✓ Validate all business rules before issuing the license.
- ✓ Prevent invalid or duplicate international licenses.
- ✓ Provide quick access to:
    - Local license information
    - License history
    - Issued international license details

Functions:

- ✓ Load local driving license using filter or passed **Local License ID**
- ✓ Validate eligibility for international license
- ✓ Automatically calculated:
    - Issue date
    - Expiration date (1 year)
    - Application fees
- ✓ Issue international driving license
- ✓ Display issued license information
- ✓ Navigate license history and license info screens

Validations:

- ✓ Only **Class 3 Local Driving Licenses** are eligible
- ✓ Only **one active International License** per driver is allowed
- ✓ International License **cannot be issued** if:
  - Local license is detained
  - Local license has expired
- ✓ Local license must be **valid and active**
- ✓ Expiration date = Issue Date + 1 year
- ✓ Application fees are retrieved from **Application Types**

Navigation:

Main Menu → Applications → Driving License Service → New Driving License → International License

---

Error Messages:

Error Code	Message	Trigger
ERR-027	Active international license already exists.	Duplicate active license
ERR-028	Invalid license class. Class 3 required.	License class is not 3
ERR-029	Local license is detained.	License is detained
ERR-030	Local license is expired.	Expiration date passed

Expected Outputs:

- ✓ International license issued successfully.
- ✓ Correct validation messages displayed.
- ✓ Accurate issue and expiration dates.
- ✓ No duplicate international licenses.
- ✓ Smooth navigation to related screens.

Notes:

- ✓ This screen contains system-driven validation only.
- ✓ No manual data entry is required from the user.
- ✓ All critical data is auto generated.

Changes Log:

Version	Date	Description of Change	Author
1.0	02 Dec 2025	Initial documentation for Issue International Driving License screen.	Abd Alrahman Saed

12. Show International Driving License Info

Document Version: **1.0**

Last Updated: **06 Nov 2026**

Author: Abd Alrahman Saed

Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **International Driving License Module**

Screen ID: **UI-IDL-012**

Screenshot:



Screen Purpose:

• Display **all details** of an issued **International Driving License** in **read-only mode**.
• Allow users to review license information without making modifications.
• Serve as a reference screen after issuing the license or from license history.

Functions:

- ✓ Load international license information by **International License ID (ILID)**
- ✓ Display related data including:
    - License holder information
    - Issued local license reference
    - Issue date
    - Expiration date
    - Application details

Navigation:

Main Menu → Manage Applications → International Licenses Application
→ Show International License Info.

Expected Outputs:

- ✓ International license information loaded successfully
- ✓ User can view all license details
- ✓ No editing options available
- ✓ Close button exits the screen correctly

Changes Log:

Version	Date	Description of Change	Author
1.0	06 Nov 2026	Initial screen documentation	Abd Alrahman Saed

13. List International Licenses Applications

Document Version: **1.0**

Last Updated: **06 Nov 2026**

Author: Abd Alrahman Saed

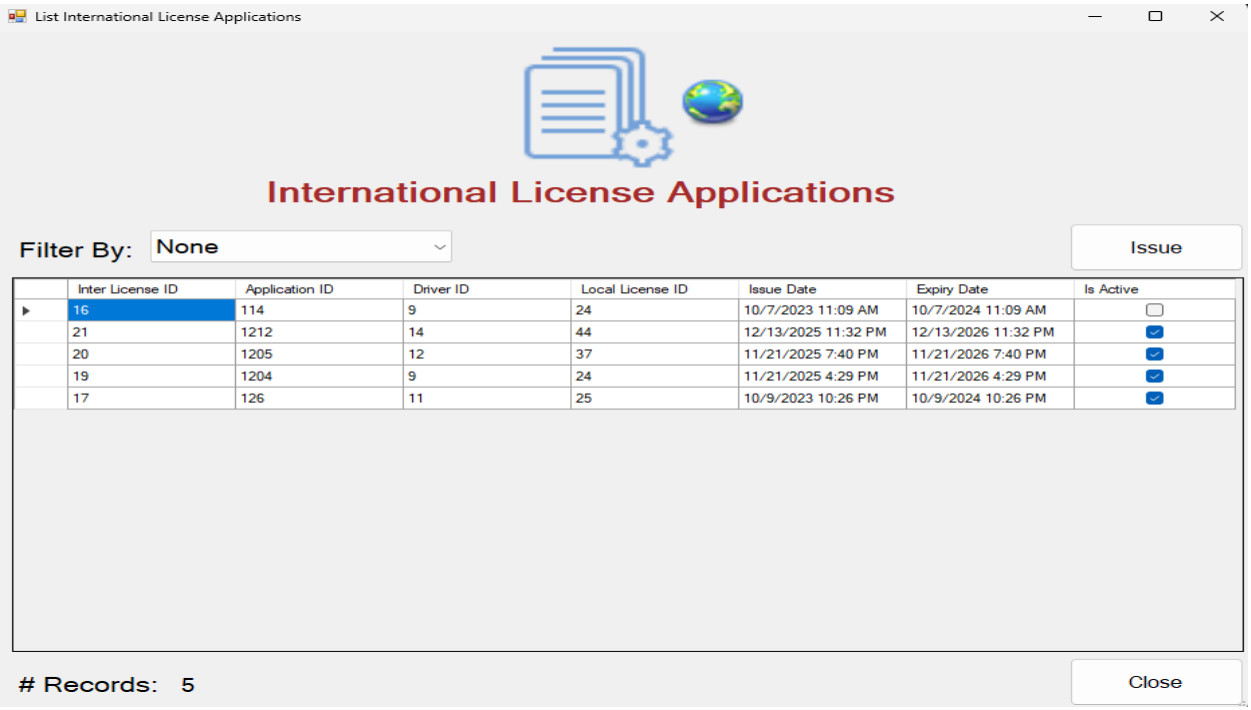
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **International Driving License Module**

Screen ID: **UI-IDL-013**

Screenshot:



Screen Purpose:

- ✓ Display a list of all issued international driving license applications.
- ✓ Allow authorized users to view and filter international license records.
- ✓ Provide access to related person, license, and history information.

Functions:

- ✓ Load all international driving licenses on screen initialization
- ✓ Display international license records in a data grid
- ✓ Show the total number of records loaded
- ✓ Issue a new international driving license via a dedicated screen
- ✓ Filter records by selected criteria:
    - International License ID
    - Application ID
    - Driver ID
    - Local License ID
    - Active Status
- ✓ Enable Active / Inactive filtering using a dropdown

Validations:

- ✓ Restrict search input to numeric values only

Navigation:

Main Menu → Manage Applications → International Licenses Application
→ Show International License Info.

Error Messages:

Error Code	Message	Trigger
ERR-061	Failed to load international licenses list.	Error retrieving data from database
ERR-062	No international license selected.	Context menu action without row selection
ERR-063	Failed to open person information screen.	Navigation failure
ERR-064	Failed to open international license screen.	Navigation failure
ERR-065	Failed to open license history screen.	Navigation failure

---

Expected Outputs:

- ✓ International license applications displayed in a structured grid
- ✓ Correct filtering and searching behavior
- ✓ Related screens opened successfully
- ✓ Updated list after issuing a new international license

Changes Log:

Version	Date	Description of Change	Author
1.0	06 Nov 2026	Initial screen documentation.	Abd Alrahman Saed

14. List Test Types

Document Version: **1.0**

Last Updated: **06 Nov 2026**

Author: Abd Alrahman Saed

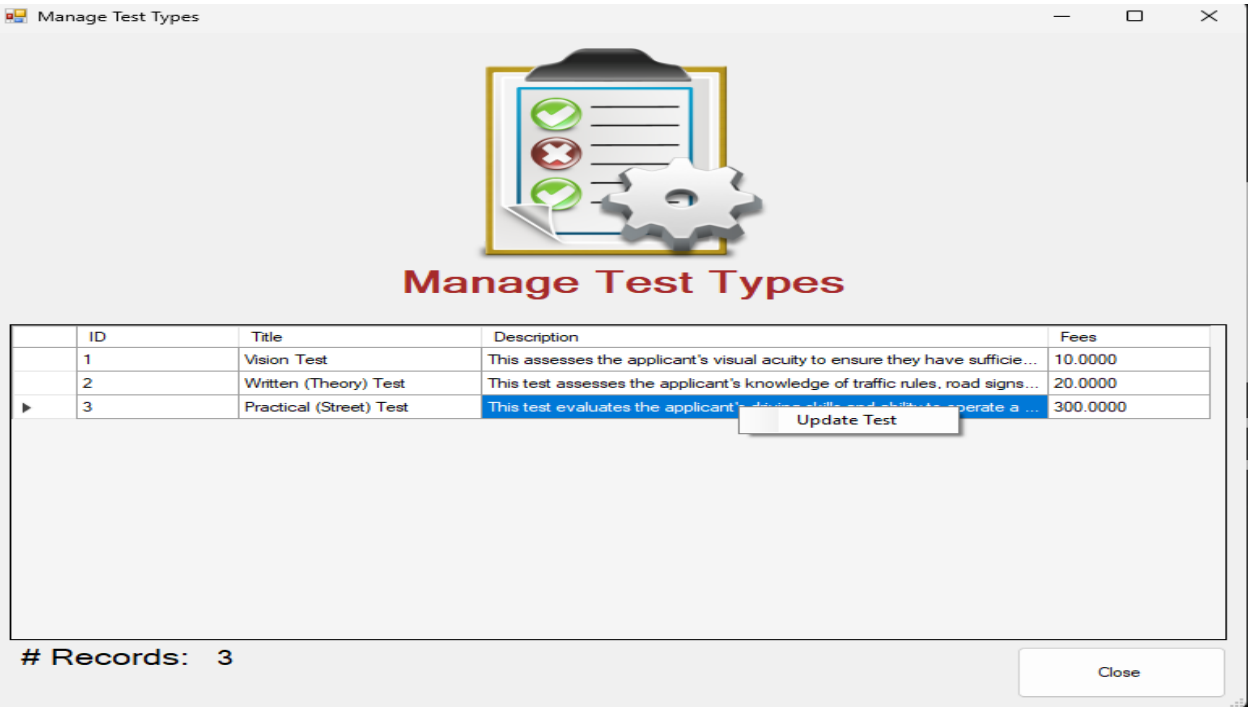
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Test Types Module**

Screen ID: **UI-TT-014**

Screenshot:



Screen Purpose:

- ✓ Display all available driving test types configured in the system.
- ✓ Allow administrative users to manage test type definitions.
- ✓ Provide a centralized reference for test titles, descriptions, and fees.

Functions:

- ✓ Load all test types from the business layer on screen load.
- ✓ Display test types in a data grid with the following columns:
    - Test ID
    - Title
    - Description
    - Fees

Validations:

- ✓ Allow updating a selected test type through a dedicated screen
- ✓ Restrict update functionality to Admin users only
- ✓ Display access denied message for non-admin users attempting updates

---

Navigation: Main Menu → Mange Test Types Module

Error Messages:

Error Code	Message	Trigger
ERR-066	Access denied. Admin privileges required.	Non-admin user attempts update action

---

Expected Outputs:

- ✓ Test types displayed correctly in the data grid
- ✓ Accurate number of records shown
- ✓ **Update screen opened only for Admin users**
- ✓ List refreshed after successful update

Changes Log:

Version	Date	Description of Change	Author
1.0	06 Nov 2026	Initial screen documentation	Abd Alrahman Saed

15. Update Test Types

Document Version: **1.0**

Last Updated: **06 Nov 2026**

Author: Abd Alrahman Saed

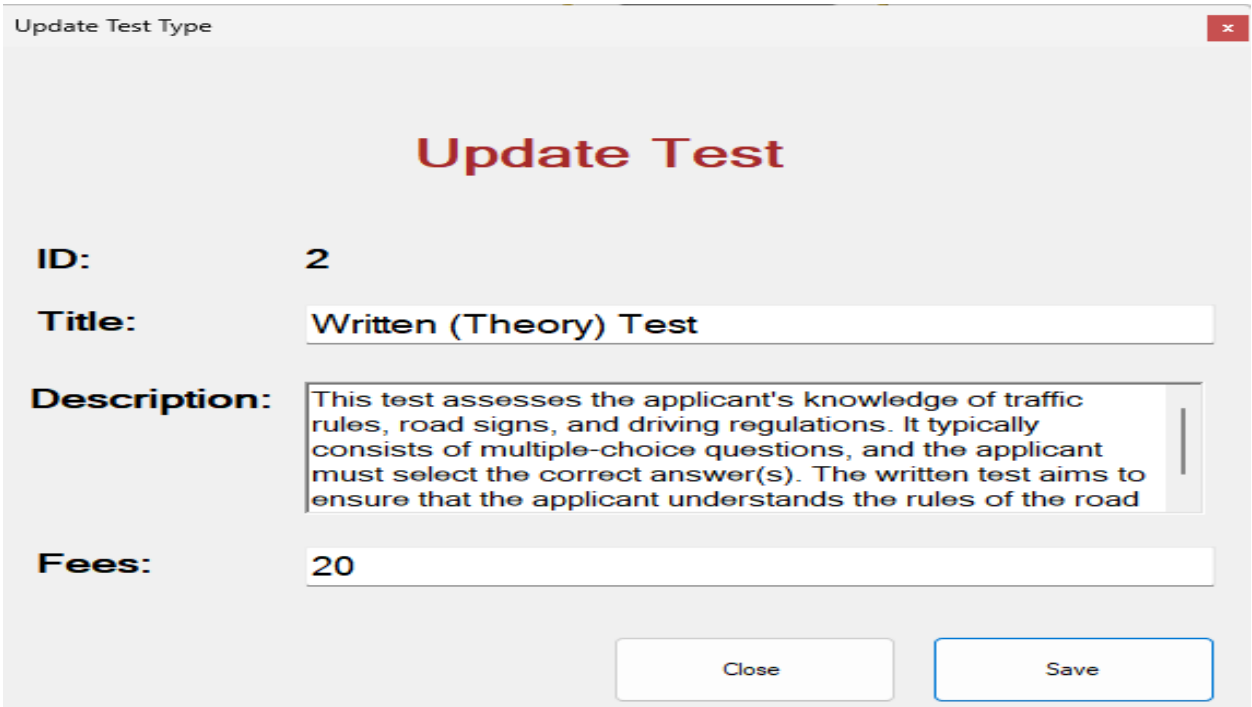
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Test Types Module**

Screen ID: **UI-TT-015**

Screenshot:



Screen Purpose:

- ✓ Allow updating existing driving test type information.
- ✓ Enable administrators to modify test title, description, and fees.
- ✓ Ensure test type data remains accurate and configurable.

Functions:

- ✓ Load selected test type details on screen load
- ✓ Display Test ID as read-only
- ✓ Populate editable fields with existing test type data

Validations:

- ✓ Validate user inputs before saving
- ✓ Restrict Fees input to numeric and decimal values only

Navigation: Main Menu → Mange Test Types Module

Error Messages:

Error Code	Message	Trigger
ERR-063	Title is required.	Title field is empty
ERR-064	Description is required.	Description field is empty
ERR-065	Please enter a valid non-negative number for fees.	Invalid or negative fees value

---

Expected Outputs:

- ✓ Updated test type data saved successfully
- ✓ Success message displayed to the user
- ✓ Updated data reflected when returning to the list screen
- ✓ Screen closes upon user action

Changes Log:

Version	Date	Description of Change	Author
1.0	06 Nov 2026	Initial screen documentation	Abd Alrahman Saed

16. Schedule Test

Document Version: **1.0**

Last Updated: **07 Nov 2026**

Author: Abd Alrahman Saed

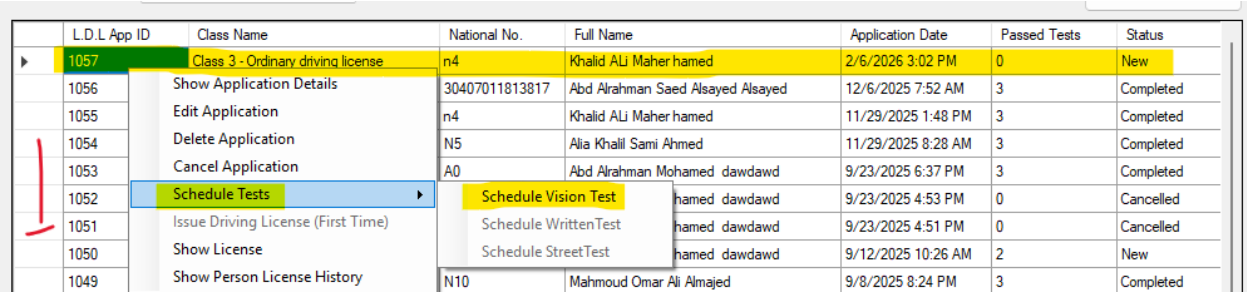
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

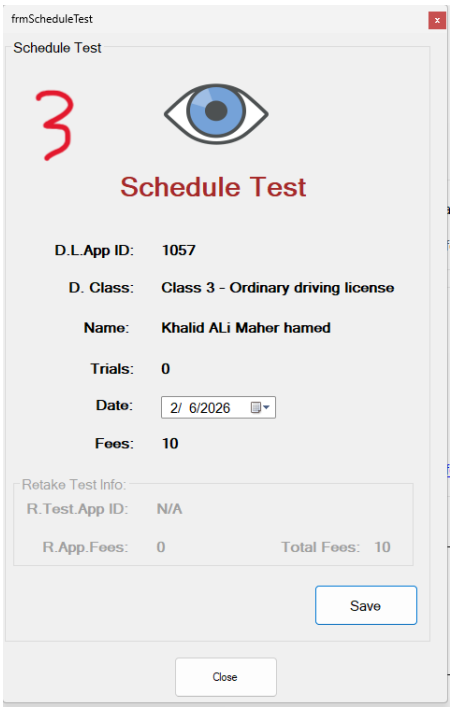
Module Name: **Tests Module**

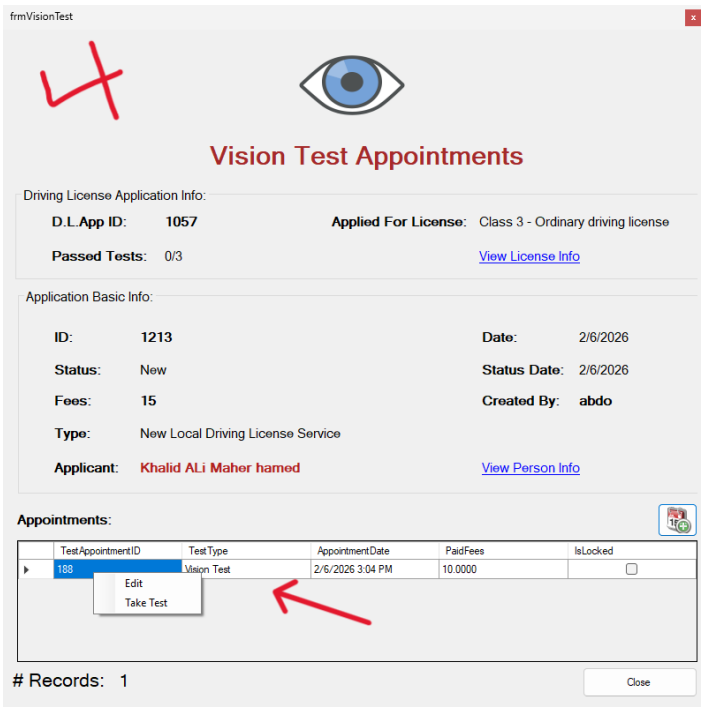
Screen ID: **UI-TST-016**

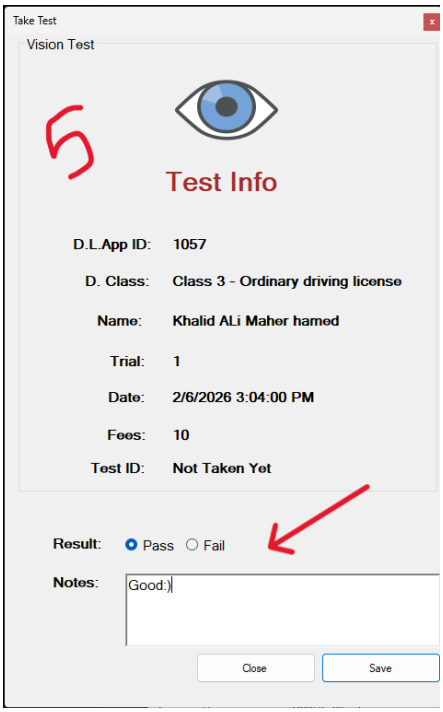
Screenshot:

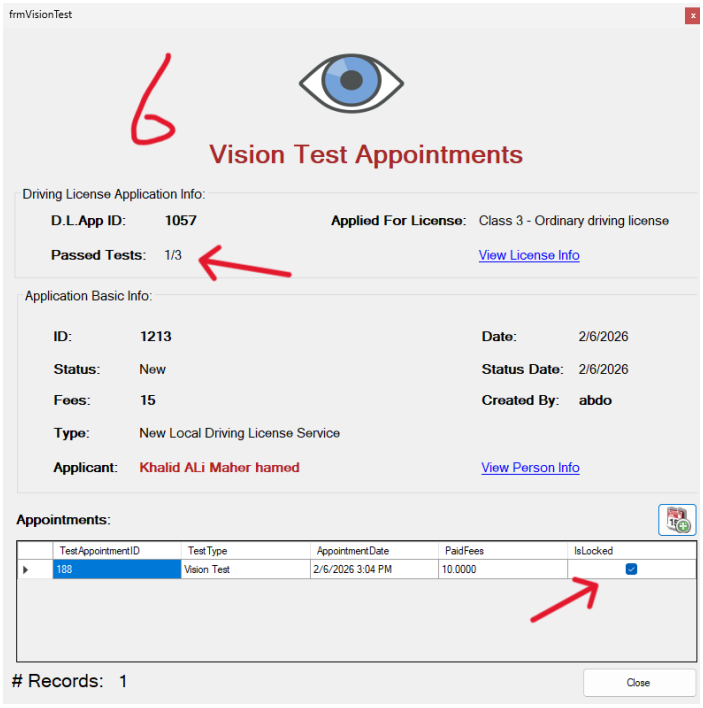


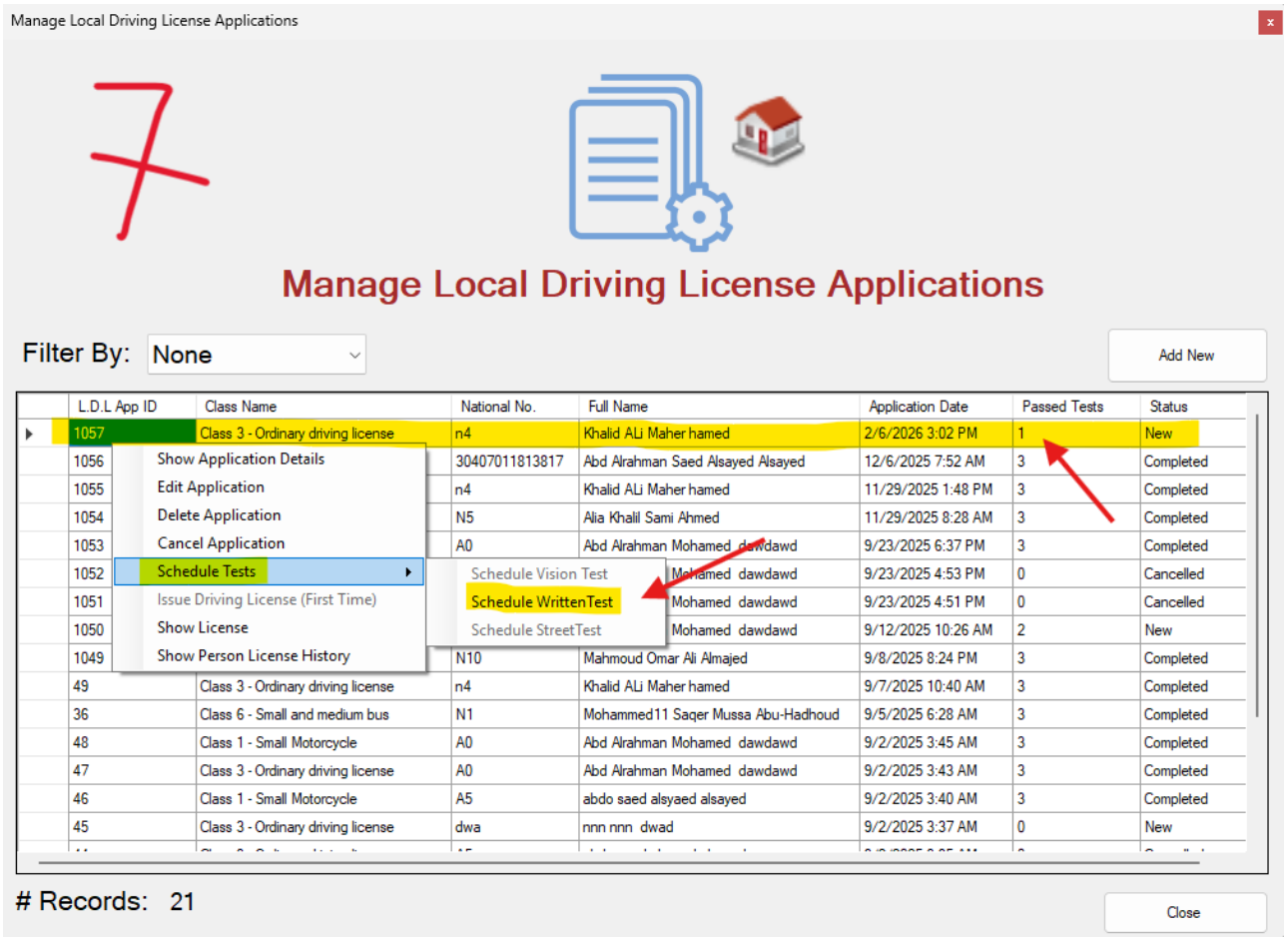



frmScheduleTest
Schedule Test
3
Schedule Test
D.L.App ID: 1057
D. Class: Class 3 - Ordinary driving license
Name: Khalid ALi Maher hamed
Trials: 0
Date: 2/ 6/2026
Fees: 10
Retake Test Info:
R.Test.App ID: N/A
R.App.Fees: 0
Total Fees: 10
Save
Close


frmVisionTest
4
Vision Test Appointments
Driving License Application Info:
D.L.App ID: 1057
Applied For License: Class 3 - Ordinary driving license
Passed Tests: 0/3
View License Info
Application Basic Info:
ID: 1213
Date: 2/6/2026
Status: New
Status Date: 2/6/2026
Fees: 15
Created By: abdo
Type: New Local Driving License Service
Applicant: Khalid ALi Maher hamed
View Person Info
Appointments:
TestAppointmentID
TestType
AppointmentDate
PaidFees
IsLocked
188
Vision Test
2/6/2026 3:04 PM
10.0000
Edit
Take Test
Records: 1
Close


Take Test
Vision Test
5
Test Info
D.L.App ID: 1057
D. Class: Class 3 - Ordinary driving license
Name: Khalid ALi Maher hamed
Trial: 1
Date: 2/6/2026 3:04:00 PM
Fees: 10
Test ID: Not Taken Yet
Result: Pass Fail
Notes: Good:)
Close
Save


frmVisionTest
6
Vision Test Appointments
Driving License Application Info:
D.L.App ID: 1057
Applied For License: Class 3 - Ordinary driving license
Passed Tests: 1/3
View License Info
Application Basic Info:
ID: 1213
Date: 2/6/2026
Status: New
Status Date: 2/6/2026
Fees: 15
Created By: abdo
Type: New Local Driving License Service
Applicant: Khalid ALi Maher hamed
View Person Info
Appointments:
TestAppointmentID
TestType
AppointmentDate
PaidFees
IsLocked
188
Vision Test
2/6/2026 3:04 PM
10.0000
Records: 1
Close



Screen Purpose:

- ✓ Schedule a driving test appointment for a local driving license application
- ✓ Handle first-time test scheduling and retake test scheduling
- ✓ Enforce test sequencing, eligibility, and appointment constraints

---

Functions:

- ✓ Display test type (Vision, Written, Practical) with corresponding title and image
- ✓ Load local driving license application information
- ✓ Display applicant details and license class
- ✓ Display number of previous trials for the selected test type
- ✓ Calculate and display test fees and retake application fees
- ✓ **Support two modes:**
  - Add New Appointment
  - Update Existing Appointment
- ✓ **Support two creation modes:**
  - First Time Test
  - Retake Test
- ✓ Save test appointment dat

Validations:

- ✓ Prevent scheduling if an active appointment exists for the same test type
- ✓ Prevent editing locked (attended) appointments
- ✓ **Enforce test order:**
- ✓ Written Test requires passing Vision Test
- ✓ Practical Test requires passing Written Test
- ✓ Automatically create and complete a Retake Test application when required
- ✓ Restrict appointment date selection based on business rules

---

Navigation:

Main Menu → Applications → Manage Applications → Local Driving License → Schedule Test

---

Error Messages:

Error Code	Message	Trigger
ERR-067	This person already has an active appointment for this test.	Active appointment exists
ERR-068	Cannot edit this appointment as it was already attended.	Appointment is locked
ERR-069	Can't schedule. You must pass Vision Test first.	Written Test prerequisite not met
ERR-070	Can't schedule. You must pass Written Test first.	Practical Test prerequisite not met
ERR-071	Failed to create retake test application.	Retake application save failed
ERR-072	Can't schedule appointment.	Test appointment save failed

---

Expected Outputs:

- ✓ Test appointment scheduled successfully
- ✓ Retake application created automatically when required
- ✓ Correct fees calculated and displayed
- ✓ Appointment date saved
- ✓ User receives confirmation message.

Changes Log:

Version	Date	Description of Change	Author
1.0	06 Nov 2026	Initial screen documentation	Abd Alrahman Saed

17. Take Test

18. List Test Appointments

The functionality, business rules, and process flow of both **Take Test** and **List Test Appointments** screens are covered and detailed in the **Schedule Test** screen documentation, since they are directly dependent on the test scheduling logic.

19. Issue License for First Time

Document Version: **1.0**

Last Updated: **08 Nov 2026**

Author: Abd Alrahman Saed

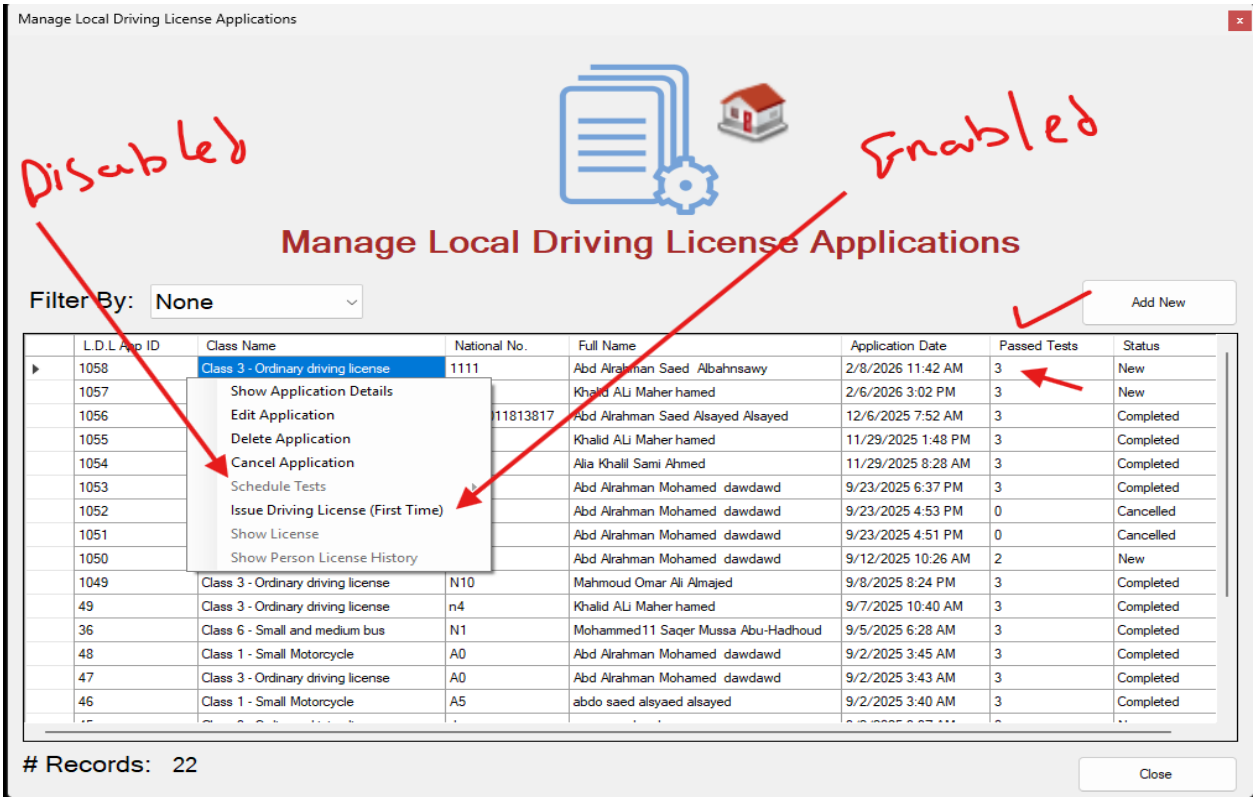
Reviewer: Dr. Heba Rashed

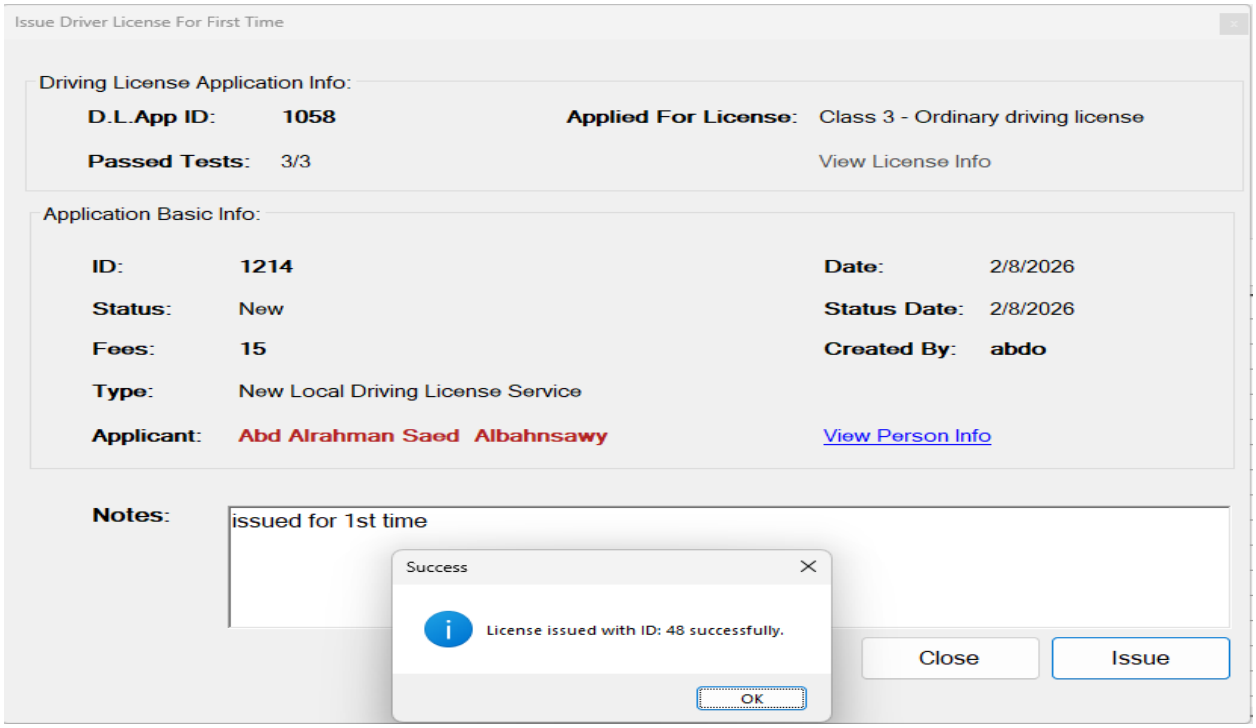
System Name: DVLD – Driving & Vehicle License Department System

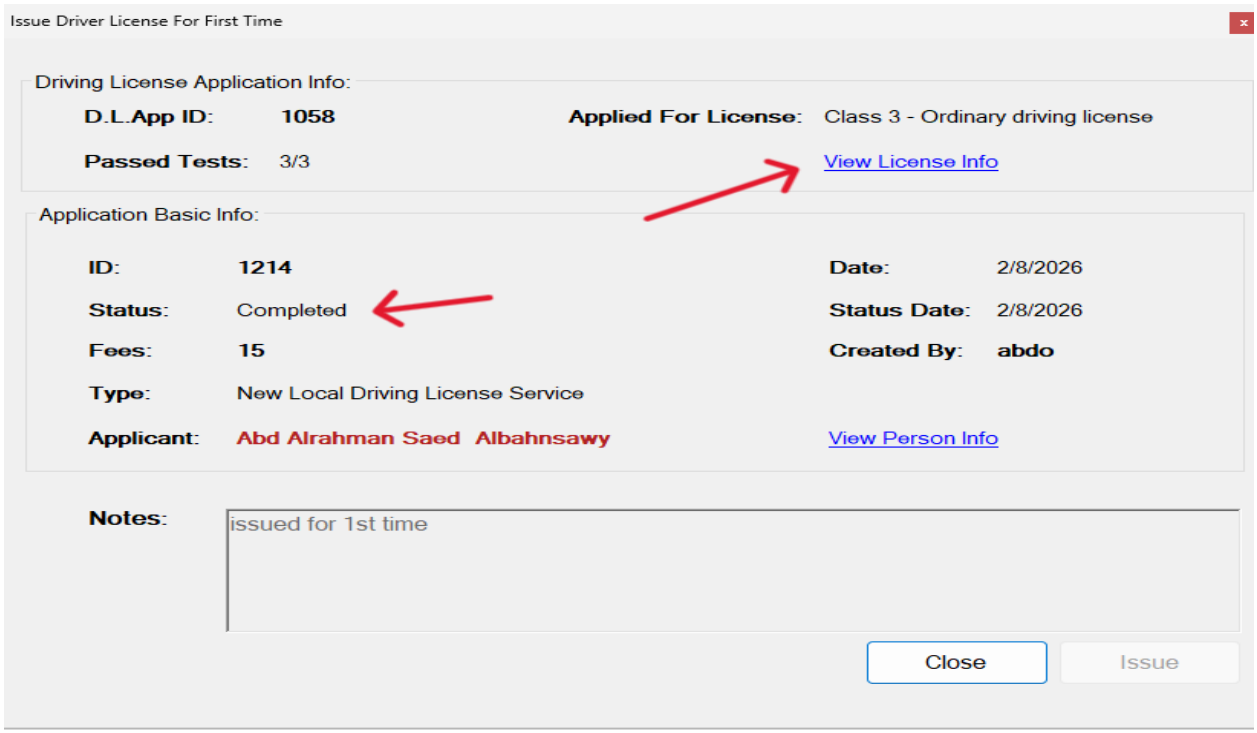
Module Name: **Local Driving Licenses Module**

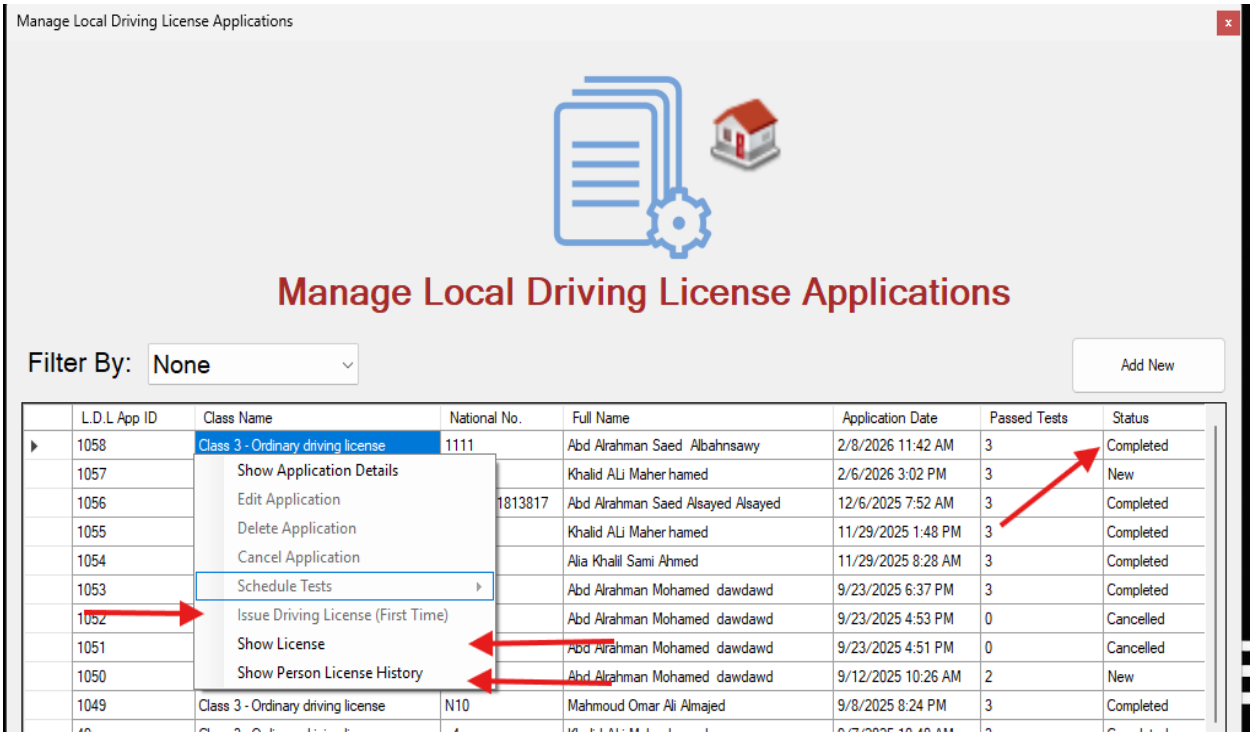
Screen ID: **UI-LDL-019**

Screenshot:




Issue Driver License For First Time
Driving License Application Info:
D.L.App ID: 1058
Applied For License: Class 3 - Ordinary driving license
Passed Tests: 3/3
View License Info
Application Basic Info:
ID: 1214
Date: 2/8/2026
Status: New
Status Date: 2/8/2026
Fees: 15
Created By: abdo
Type: New Local Driving License Service
Applicant: Abd Alrahman Saed Albahnsawy
View Person Info
Notes:
issued for 1st time
Success
License issued with ID: 48 successfully.
OK
Close
Issue


Issue Driver License For First Time
Driving License Application Info:
D.L.App ID: 1058
Applied For License: Class 3 - Ordinary driving license
Passed Tests: 3/3
View License Info
Application Basic Info:
ID: 1214
Date: 2/8/2026
Status: Completed
Status Date: 2/8/2026
Fees: 15
Created By: abdo
Type: New Local Driving License Service
Applicant: Abd Alrahman Saed Albahnsawy
View Person Info
Notes:
issued for 1st time
Close
Issue



Screen Purpose:

- ✓ Issuing a local driving license for the first time.
- ✓ Validate eligibility of the local driving license application before issuance.
- ✓ Finalize the licensing process after successful completion of all required tests.

Functions:

- ✓ Load local driving license application details
- ✓ Validate existence of the application
- ✓ Verify that all required tests are passed
- ✓ Check that no active license already exists for the application
- ✓ Allow entering optional notes during license issuance
- ✓ Issuing a new driving license for the first time
- ✓ Displaying newly issued license information
- ✓ Disable issue action after successful issuance

Validations:

- ✓ Validates application existence
- ✓ Verifies test completion status
- ✓ Checks for existing active license

Navigation:

Main Menu → Applications → Manage Applications → Local Driving License →

Issue License for first time

Error Messages:

Error Code	Message	Trigger
ERR-073	All required tests are not passed.	Tests not completed
ERR-074	This person already has an active license.	Active license already exists

---

Expected Outputs:

- ✓ New local driving license issued successfully
- ✓ License ID displayed to the user
- ✓ License information shown on the application card
- ✓ Issue action disabled after completion
- ✓ Updated Application Status to be completed
- ✓ Disable Issue for first time button
- ✓ Enabled Show License info and history buttons

Changes Log:

Version	Date	Description of Change	Author
1.0	08 Nov 2026	Initial screen documentation	Abd Alrahman Saed

20. Show Driver License Info

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Local Driving Licenses Module**

Screen ID: **UI-LDL-020**

Screenshot:



Screen Purpose:

- ✓ Display detailed information about a specific local driving license
- ✓ Provide a read-only view of issued license data

Functions:

- ✓ Receive License ID from calling screenEdit
- ✓ Load driver license information using the License ID
- ✓ Display license details through the Driver License Information control

---

Validations: No validation on this screen.

Navigation:

Main Menu → Applications → Manage Applications → Local Driving License →

Show License Info

Error Messages:

Error Code	Message	Trigger
ERR-075	Failed to load license information.	Failed to load license information.

Expected Outputs:

- Driver license information displayed successfully.
- User able to review full license details.

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

21. Show Driver License History

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

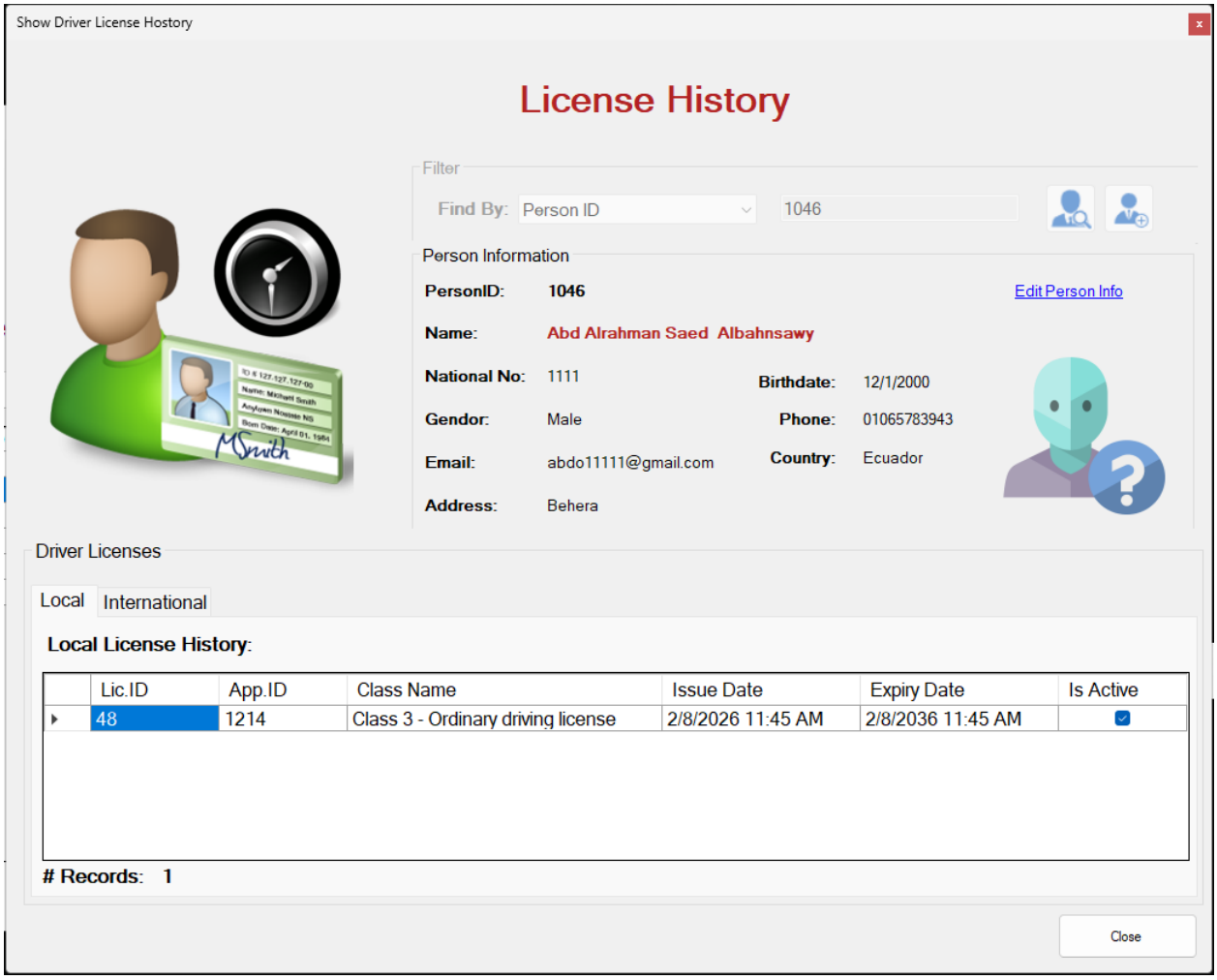
Reviewer: Dr. Heba Rashed

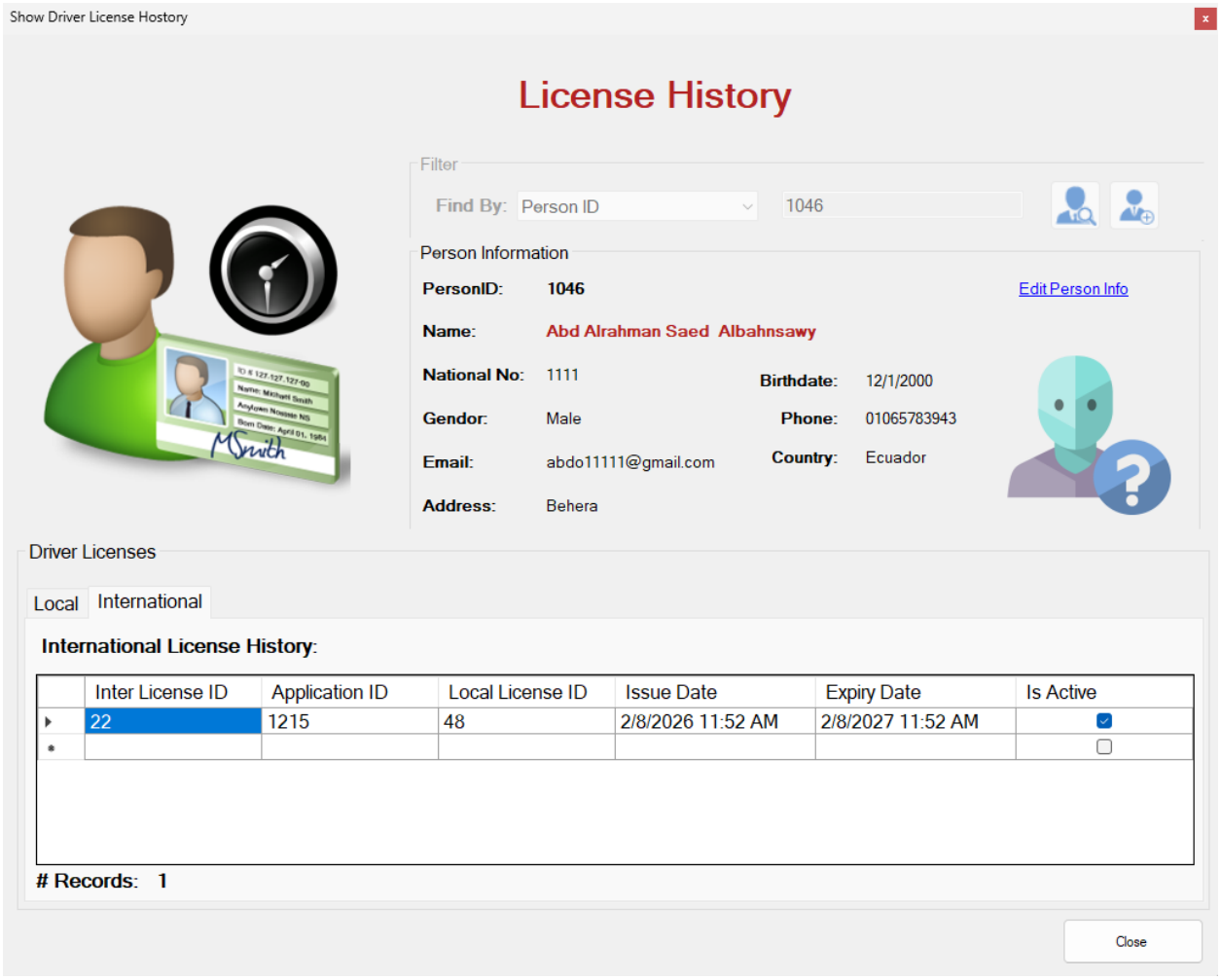
System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Licenses Module**

Screen ID: **UI-LCNS-003**

Screenshot:





Screen Purpose:

- ✓ Display the complete license history for a specific person
- ✓ Provide a consolidated view of all driving licenses issued to a driver
- ✓ Allow searching and selecting a person to view their license records.

Functions:

- ✓ Allow selecting a person using a filter control when no Person ID is provided
- ✓ Display selected person information
- ✓ Display all licenses associated with the selected person
- ✓ Automatically disable person filter when Person ID is predefined
- ✓ Screen is read-only

Validations:

- ✓ Person ID or National No. must be valid to load license history
- ✓ When Person ID is predefined, filter is disabled
- ✓ When Person ID is not predefined, filter is enabled

Navigation:

Main Menu → Applications → Manage Applications → Local Driving License →

Show Licenses History

Error Messages:

Error Code	Message	Trigger
ERR-076	Person not found.	Invalid Person ID or National Number
ERR-077	Failed to load person license history.	Database error while loading history

---

Expected Outputs:

- Person information displayed correctly
- Complete list of local licenses displayed
- Filter enabled when needed
- License history cleared when no valid person is selected

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

22. List Drivers

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

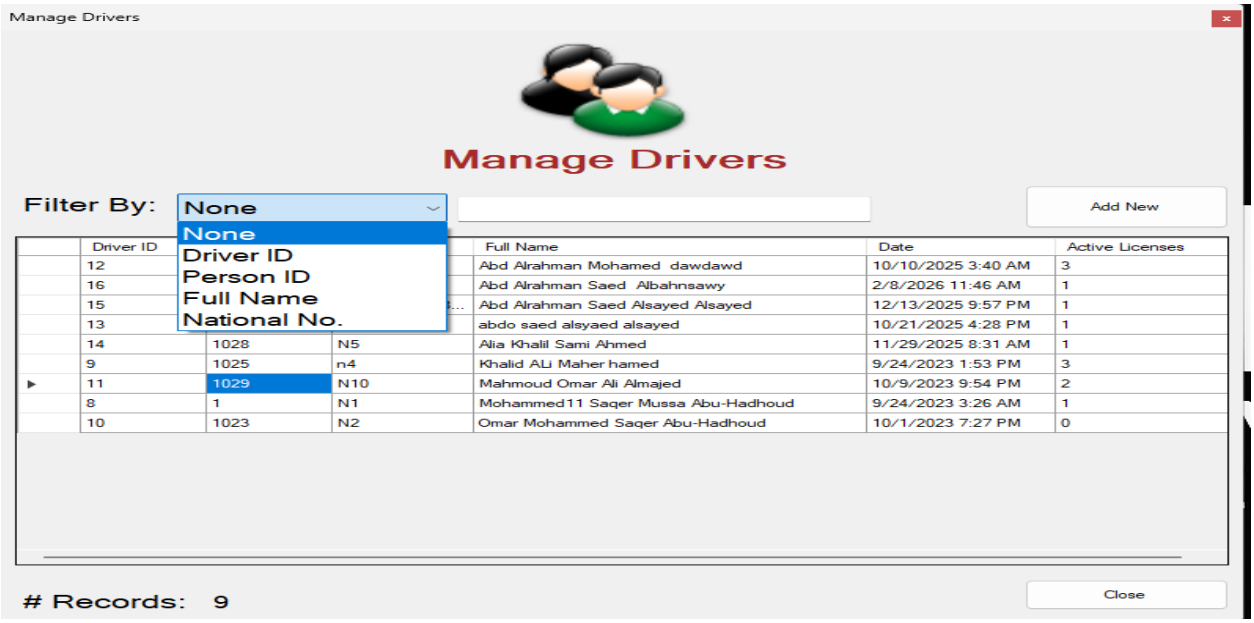
Reviewer: Dr. Heba Rashed

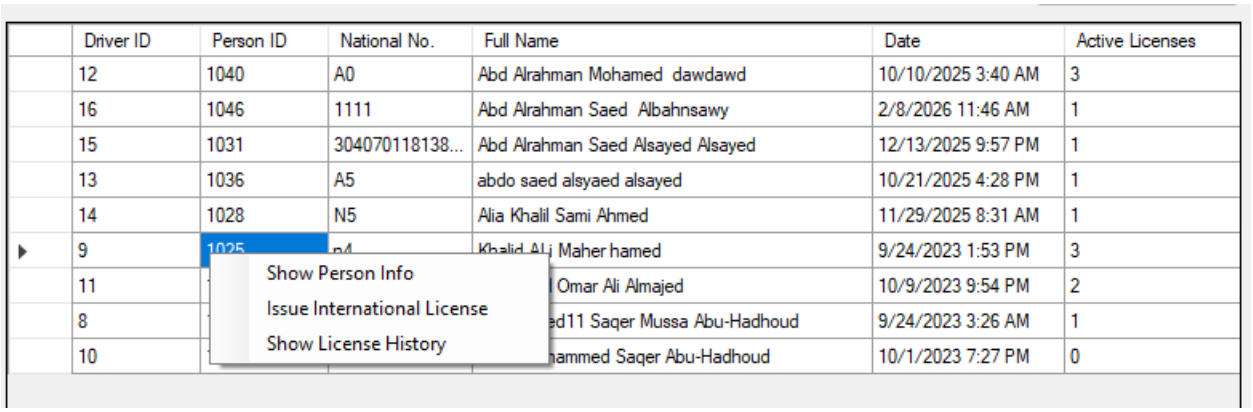
System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Drivers Module**

Screen ID: **UI-DRV-022**

Screenshot:





Screen Purpose:

- ✓ Display all registered drivers in the system
- ✓ Allow searching and filtering driver records
- ✓ Provide quick navigation to related drivers and license information

---

Functions:

- ✓ Load all drivers registered in the system
- ✓ Display total number of records
- ✓ Filter drivers by:
  - Driver ID
  - Person ID
  - Full Name
  - National Number
- ✓ Support partial search for Full Name and National Number
- ✓ Open Add New Local Driving License Application screen
- ✓ Open Show Person Information screen
- ✓ Open Show Driver License History screen

Validations:

- ✓ Restrict numeric input for Driver ID and Person ID filters

---

Navigation: Main Menu → Drivers

Error Messages:

Error Code	Message	Trigger
ERR-078	Failed to load drivers list.	Data retrieval failure

Expected Outputs:

- ✓ Drivers displayed correctly in data grid
- ✓ Accurate record count shown
- ✓ Filtering updates results dynamically
- ✓ Related screens open successfully

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2025	Initial screen documentation	Abd Alrahman Saed

23. Detain License

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

Reviewer: Dr. Heba Rashed

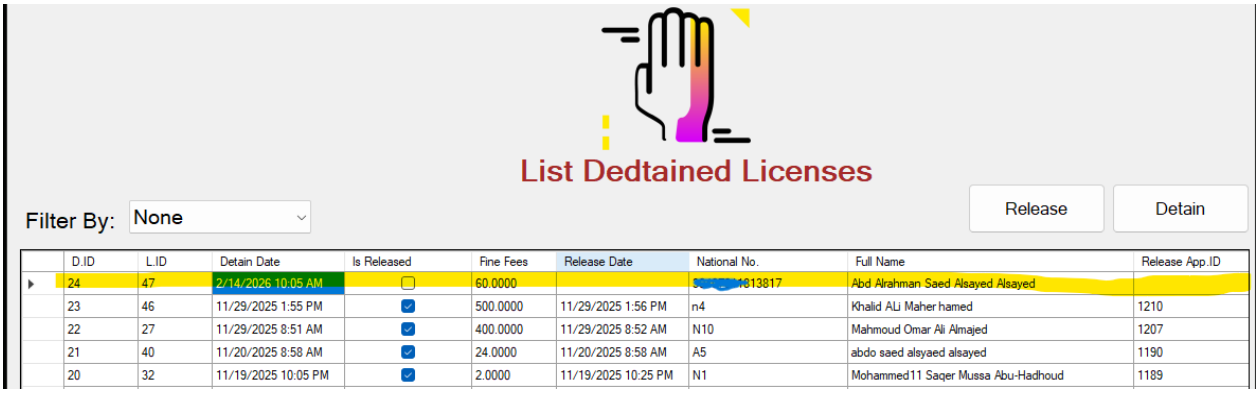
System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Detained Licenses Module**

Screen ID: **UI-DTL-023**

Screenshot:





Screen Purpose:

- ✓ Detain an active local driving license
- ✓ Record fine fees associated with license detention
- ✓ Provide access to related license information and license history

---

Functions:

- ✓ Select a license using Driver License filter control
- ✓ Display selected license information
- ✓ Display detained license details if already detained
- ✓ Confirm detention action before processing
- ✓ Disable further modifications after successful detention
- ✓ Open Driver License History screen
- ✓ Open Driver License Information screen

Validations:

- ✓ Check if license is already detained
- ✓ Validate fine fees input

Navigation: Main Menu → Detained Licenses → Detain License

Error Messages:

Error Code	Message	Trigger
ERR-079	Fine Fees is required.	Fine Fees field empty
ERR-080	Fine Fees must be a valid non-negative number.	Invalid fine fees input
ERR-081	This license is already detained.	Attempt to detain already detained license

Error Code	Message	Trigger
ERR-082	Error detaining license.	Detain operation failed

Expected Outputs:

- License successfully detained.
- Detained License ID generated and displayed
- Fine fees recorded
- Related license screens accessible

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2025	Initial screen documentation	Abd Alrahman Saed

24. List Detained Licenses

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

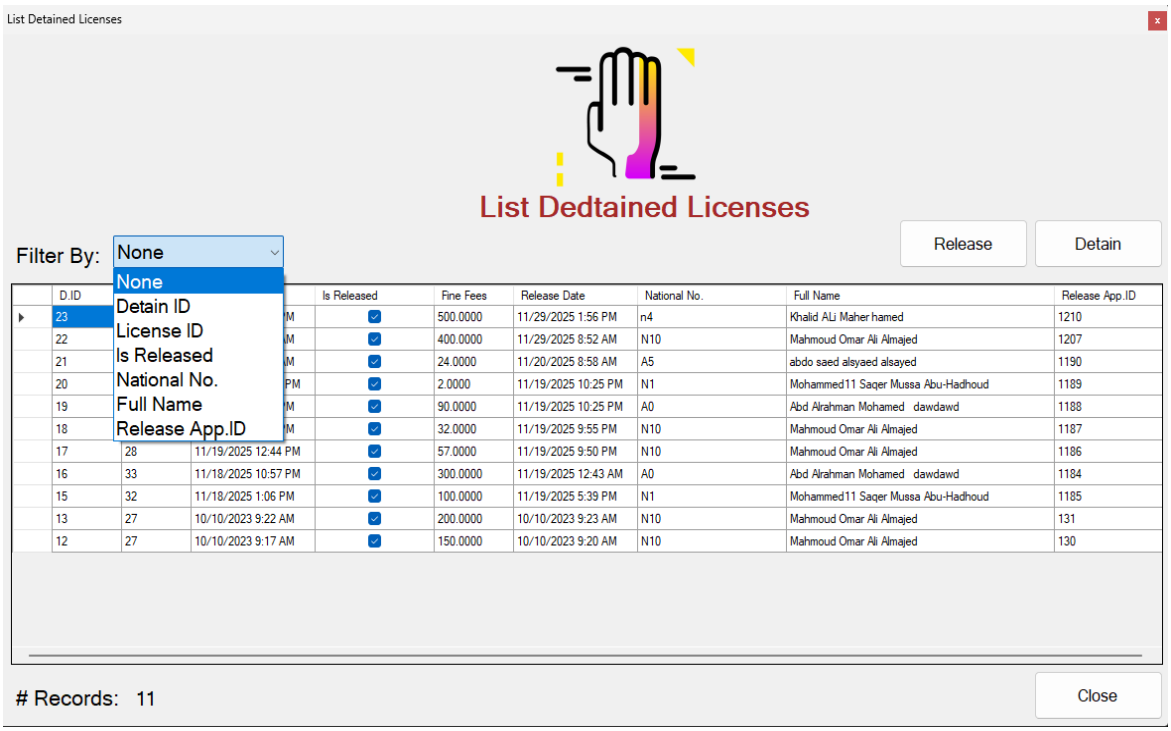
Reviewer: Dr. Heba Rashed

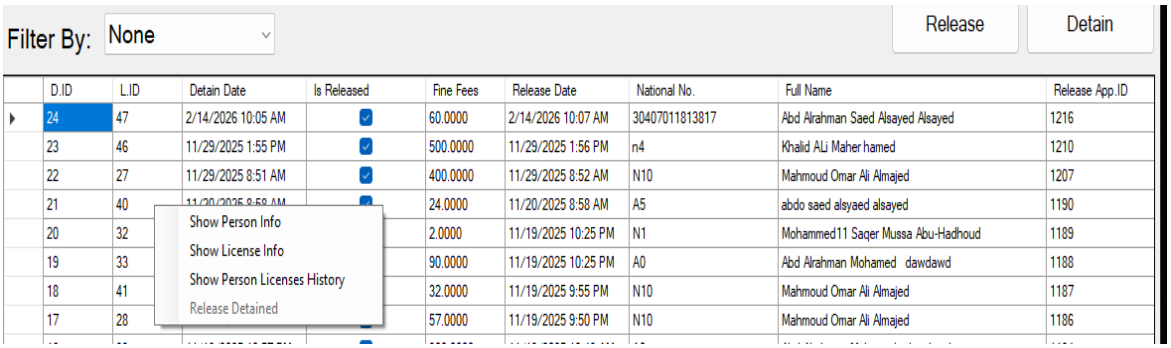
System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Detained Licenses Module**

Screen ID: **UI-DTL-024**

Screenshot:





Screen Purpose:

- ✓ Display all detained licenses in the system
- ✓ Allow filtering and searching detained license records
- ✓ Provide access to detain and release operations
- ✓ Enable navigation to related person and license details

Functions:

- ✓ Load all detained licenses in the system
- ✓ Filter records by:
  - Detain ID
  - License ID
  - Is Released
  - National Number
  - Full Name
  - Release Application ID
- ✓ Open Show Person Information screen
- ✓ Open Show Driver License Information screen
- ✓ Open Show Driver License History screen

Validations:

- ✓ Restrict numeric input for ID-based filters
- ✓ Support partial matching for National Number and Full Name

---

Navigation: Main Menu → Detained Licenses → Manage Detained

Error Messages:

Error Code	Message	Trigger
ERR-083	Failed to load detained licenses list.	Data retrieval failure

---

Expected Outputs:

- ✓ Detained licenses displayed correctly
- ✓ Accurate record count shown
- ✓ Filtering updates results dynamically
- ✓ Detain and release screens open successfully
- ✓ Release option disabled when record is already released
- ✓ List refreshed after operations.

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

25. Release Detained License

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

Reviewer: Dr. Heba Rashed

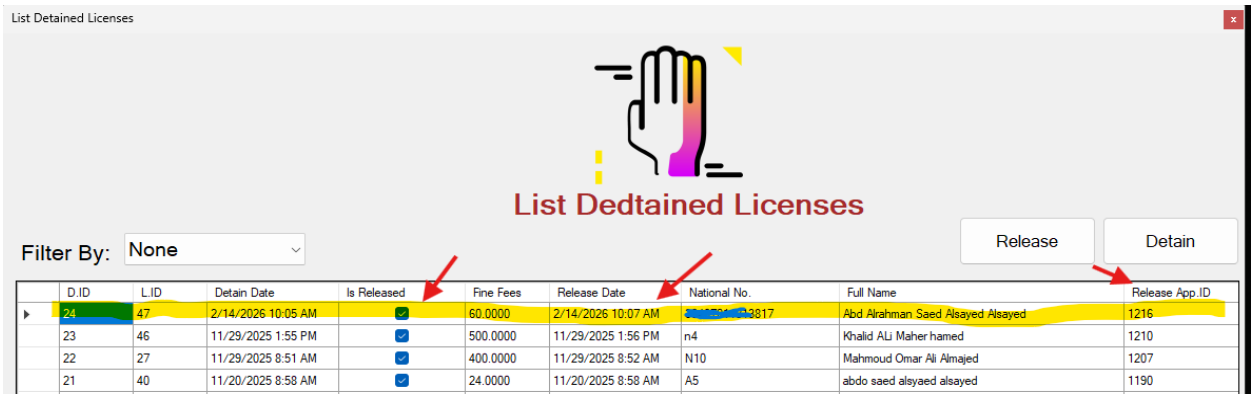
System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Release Detained LicenseModule**

Screen ID: **UI-RDL-025**

Screenshot:





Screen Purpose:

- ✓ Release a previously detained local driving license
- ✓ Calculate and display total payable fees before release
- ✓ Generate a release application linked to the detention record

---

Functions:

- ✓ Select a license using Driver License filter control
- ✓ Open Driver License Information screen
- ✓ Open Driver License History screen
- ✓ Release detained license
- ✓ Generate Release Application ID
- ✓ Calculate total fees (Fine Fees + Release Application Fees)
- ✓ Disable release button after successful operation

Validations:

- ✓ Validate that the selected license is currently detained

Navigation: Main Menu → Detained Licenses → Release Detained licenses

---

Error Messages:

Error Code	Message	Trigger
ERR-084	This license is not detained.	Attempt to release non-detained license
ERR-085	Failed to release detained license.	Release operation failed

---

Expected Outputs:

- ✓ License successfully released
- ✓ Release Application ID generated and displayed
- ✓ Updated license status reflected in system
- ✓ Confirmation message shown
- ✓ Related screens accessible

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

26. Renew Local Driving License

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

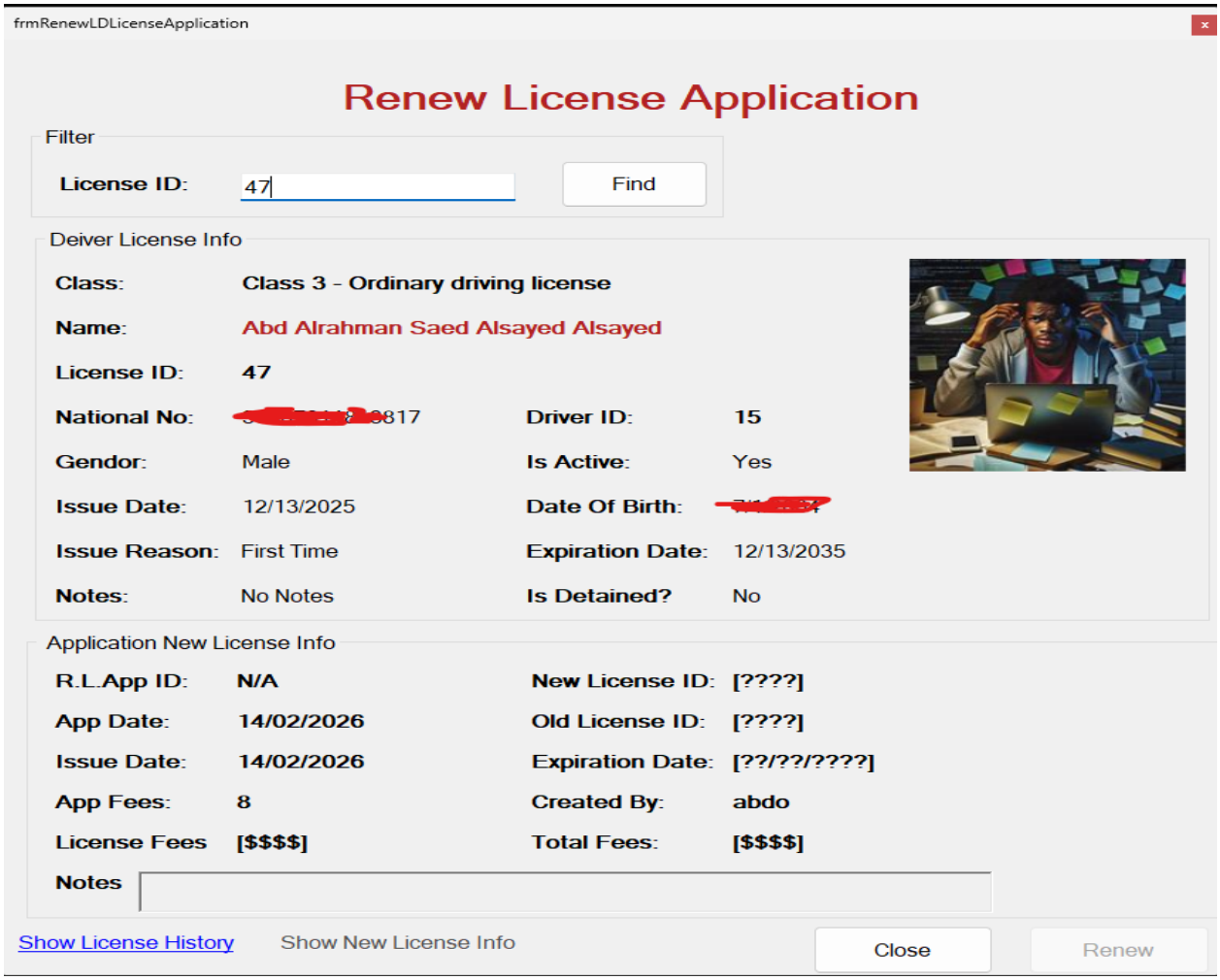
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Renew Local Driving License Module**

Screen ID: **UI-RLL-026**

Screenshot:



Screen Purpose:

- ✓ Renew an expired and active local driving license
- ✓ Generate a renewal application and a new license record
- ✓ Calculate renewal fees and determine new expiration date

Functions:

- ✓ Select license using Driver License filter control
- ✓ Display All License Info
- ✓ Delete
- ✓ Search
- ✓ Issue License
- ✓ Calculate new expiration date based on license class default validity
- ✓ Calculate renewal fees and determine new expiration date
- ✓ Allow updating notes
- ✓ Confirm renewal operation
- ✓ Generate new license and renewal application
- ✓ Display:
  - New License ID
  - Renewal Application ID
- ✓ Disable renewal action after success
- ✓ Open new license information screen

Validations:

- ✓ Validate selected license
- ✓ Check if license is expired
- ✓ Check if license is active

Navigation: Main Menu → Driving Licenses Services → Renew Driving License

---

Error Messages:

Error Code	Message	Trigger
ERR-086	The selected license is not expired.	Attempt to renew non-expired license
ERR-087	The selected license is not active.	Attempt to renew inactive license
ERR-088	An error occurred while renewing the license.	Renewal operation failed

---

Expected Outputs:

- ✓ License renewed successfully
- ✓ New License ID generated and displayed
- ✓ Renewal Application ID displayed
- ✓ Confirmation message shown
- ✓ New license information accessible
- ✓ Renewal action disabled after completion

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

27. Replace Lost or Damaged License

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

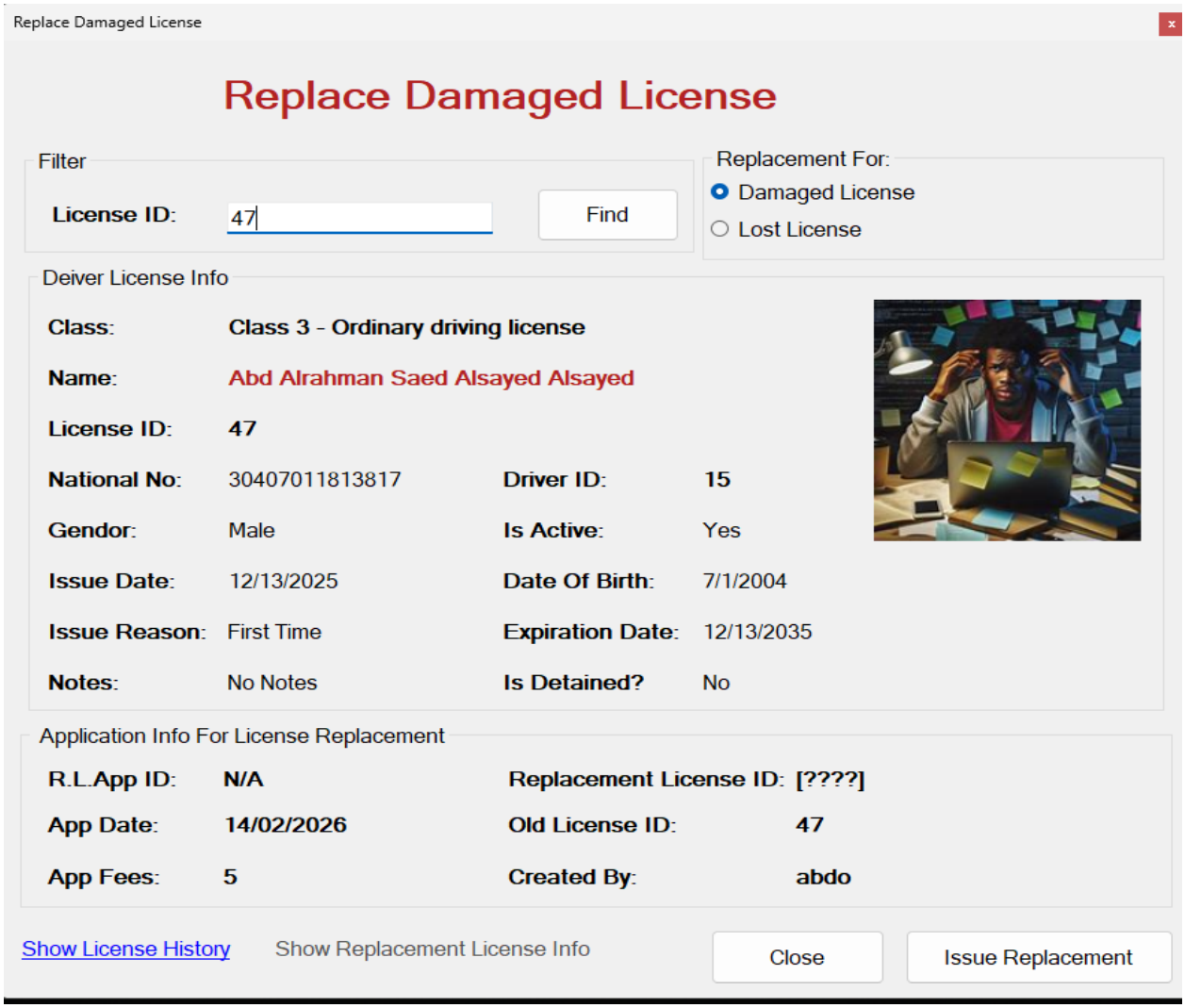
Reviewer: Dr. Heba Rashed

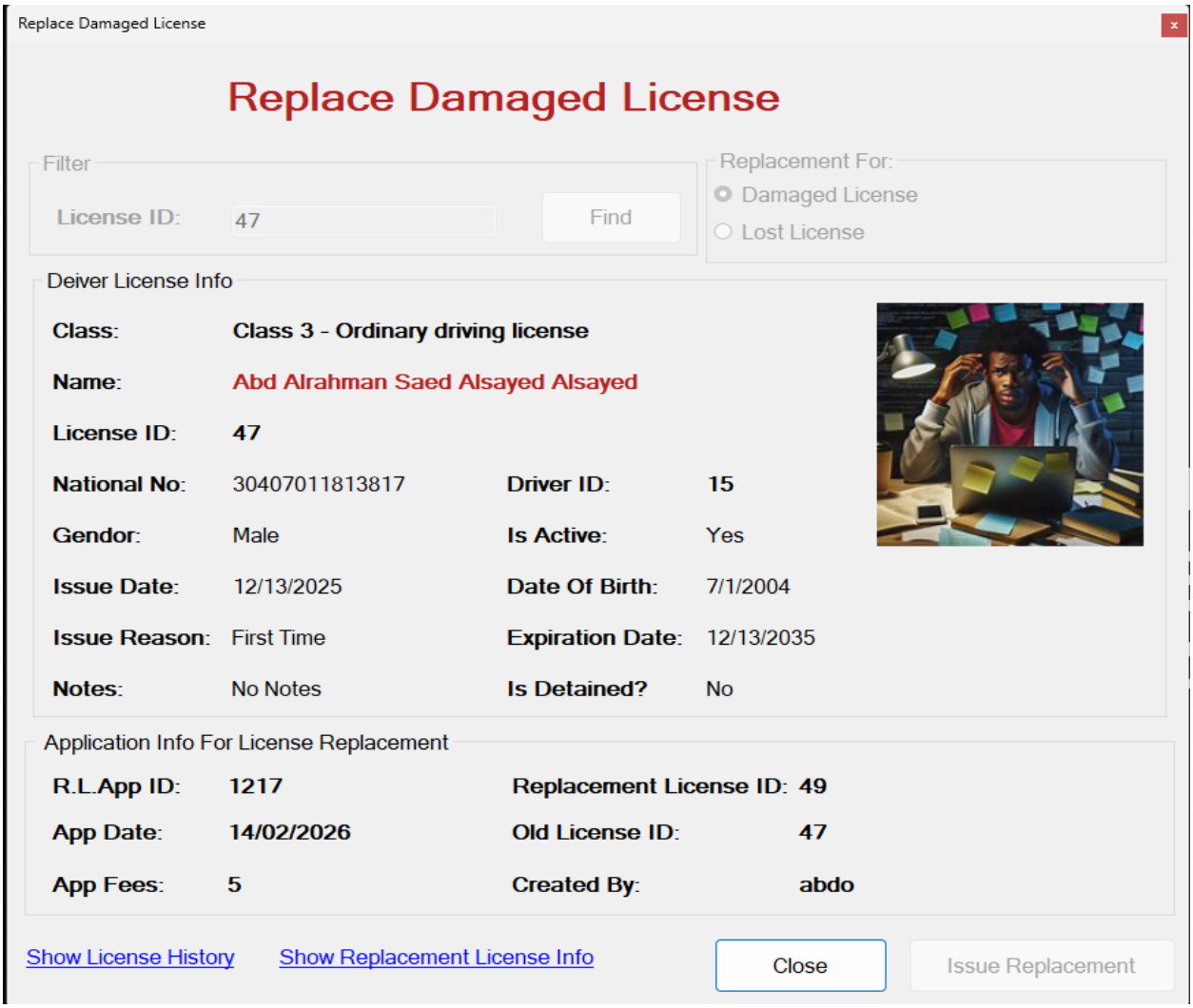
System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Replace Lost or Damaged License Module**

Screen ID: **UI-RPL-027**

Screenshot:





Screen Purpose:

- ✓ Issue a replacement license for a lost or damaged local driving license
- ✓ Generate a replacement application and new license record
- ✓ Record the issue reason (Lost or Damaged)

Functions:

- ✓ Select license using Driver License filter control
- ✓ Display All License Info:
- ✓ Lost License
- ✓ Damaged License
- ✓ Select replacement reason:
- ✓ Dynamically update:
    - Application Fees
    - Screen header and title
- ✓ Confirm replacement operation
- ✓ Replacement Application
- ✓ Disable replacement after successful issuance
- ✓ Open new license information screen

Validations:

- ✓ Validate selected license

Navigation:

Main Menu → Driving Licenses Services → Replace Lost or Damaged Driving License

---

Error Messages:

Error Code	Message	Trigger
ERR-089	The selected license is not active.	Attempt to replace inactive license
ERR-090	An error occurred while issuing the replacement license.	Replacement operation failed

---

Expected Outputs:

- ✓ Replacement license issued successfully
- ✓ Replacement License ID generated and displayed
- ✓ Replacement Application ID displayed
- ✓ Confirmation message shown
- ✓ New license information accessible
- ✓ Replacement process locked after completion

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

28. Add / Edit User Information

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

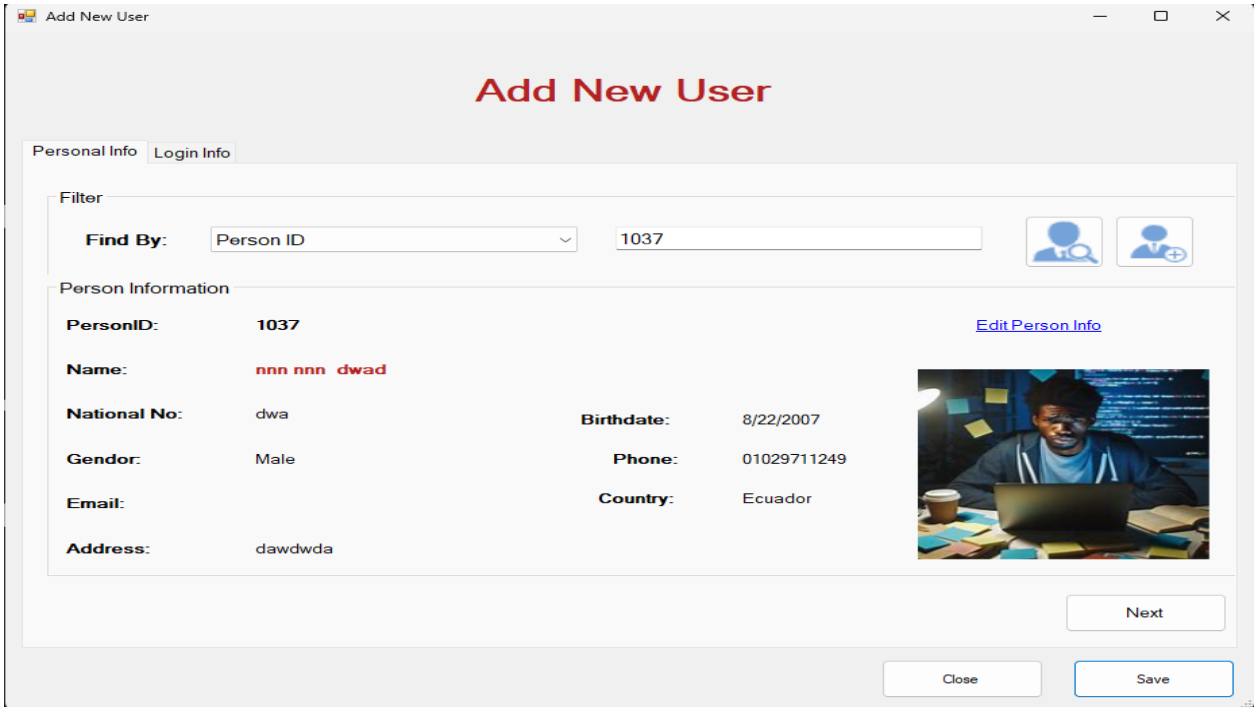
Reviewer: Dr. Heba Rashed

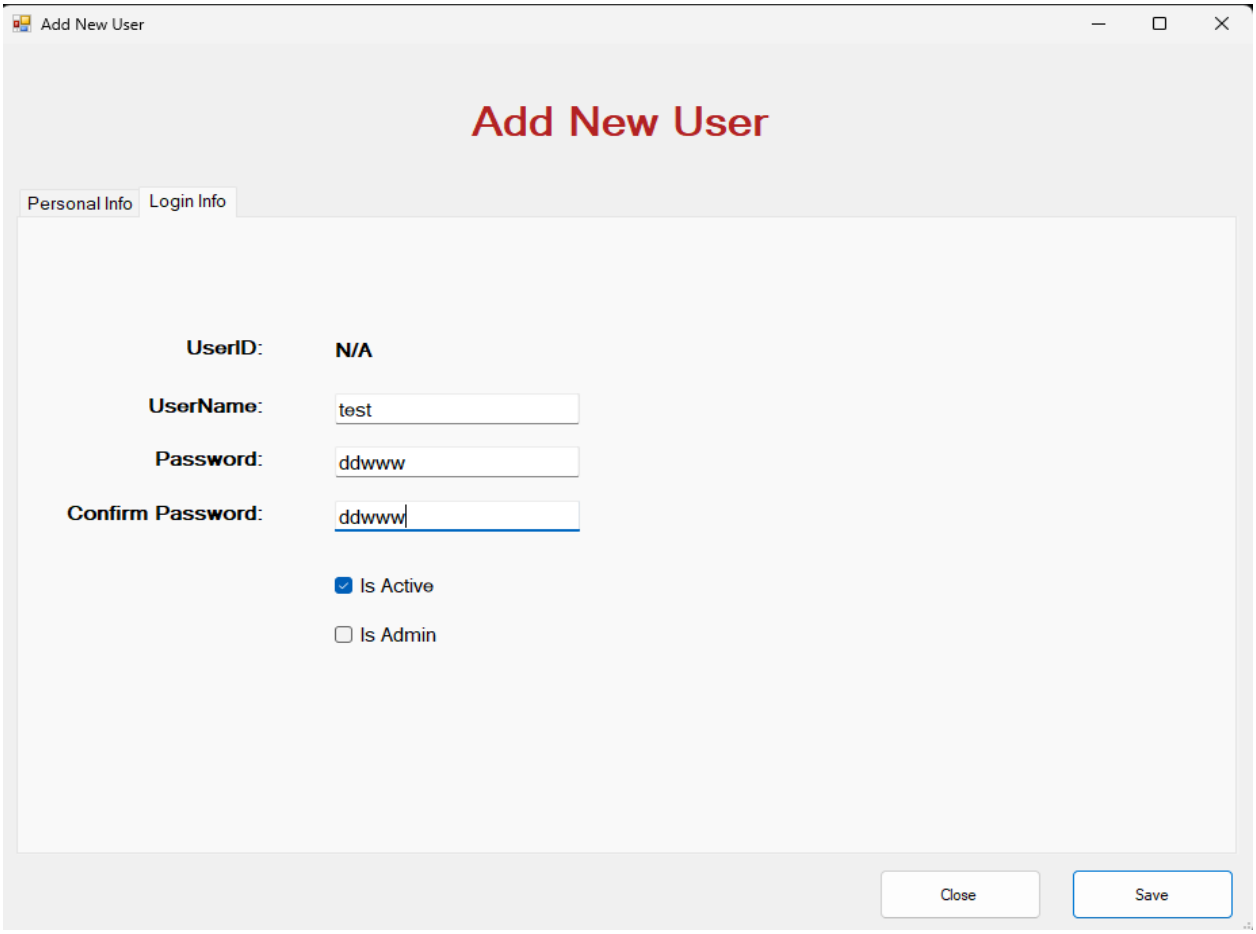
System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Users Module**

Screen ID: **UI-USR-028**

Screenshot:





Screen Purpose:

- ✓ Add a new system user
- ✓ Edit existing user information
- ✓ Link user accounts to existing people
- ✓ Manage login credentials and user privileges

Functions:

- ✓ Support two modes:
    - Add New User
    - Update Existing User
- ✓ Save new user or update existing user
- ✓ Automatically switch from Add mode to Update mode after successful save

---

Validations:

- ✓ Validate required fields
- ✓ Validate password confirmation
- ✓ Prevent creating duplicate user accounts for the same person

---

Navigation:

Main Menu → Users → Add New User

Main Menu → Users → Edit User

Error Messages:

Error Code	Message	Trigger
ERR-091	This user already exists.	Duplicate user for same PersonID
ERR-092	Please select a person first.	No person selected in Add mode
ERR-093	Username is required.	Empty username field
ERR-094	Password is required.	Empty password field
ERR-095	Confirm password does not match.	Password confirmation mismatch
ERR-096	Error saving user information.	Save operation failed
ERR-097	Error loading user information.	Load operation failed

Expected Outputs:

- ✓ New user created successfully
- ✓ Existing users updated successfully
- ✓ User ID displayed after creation
- ✓ User privileges saved correctly
- ✓ Confirmation message shown

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

29. Show User Information

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

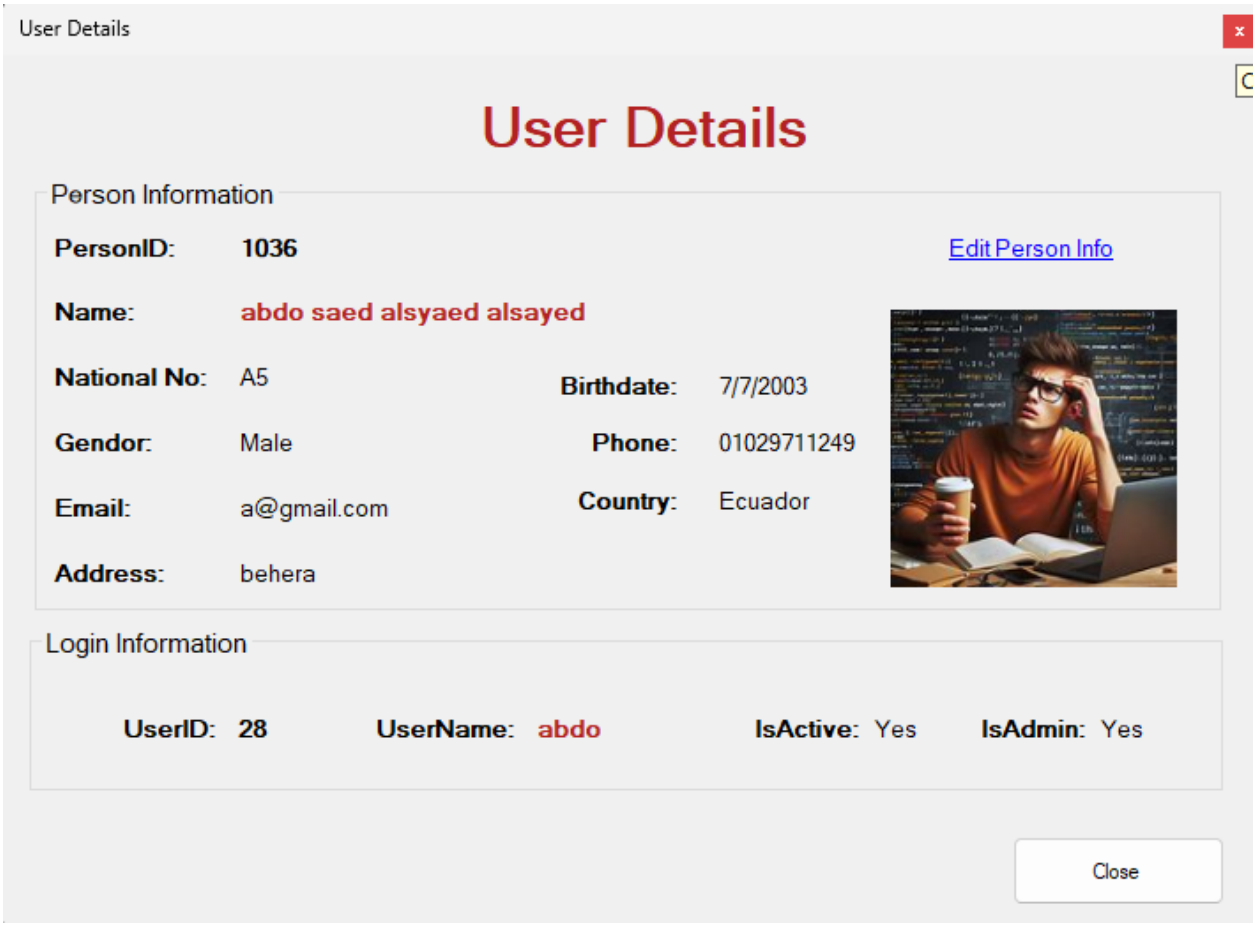
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Users Module**

Screen ID: **UI-USR-029**

Screenshot:



Screen Purpose:

Display detailed information about a specific system user.

Functions:

- ✓ Load user information using User ID
- ✓ Display user details through User Card control

Validations:

- ✓ Name required
- ✓ Fees numeric
- ✓ Driver must exist

Navigation: Main Menu → Users → Show User Information

Error Messages:

Error Code	Message	Trigger
ERR-098	Failed to load user information.	User data load failure

---

Expected Outputs:

- ✓ User information displayed successfully.

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

30. List Users

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

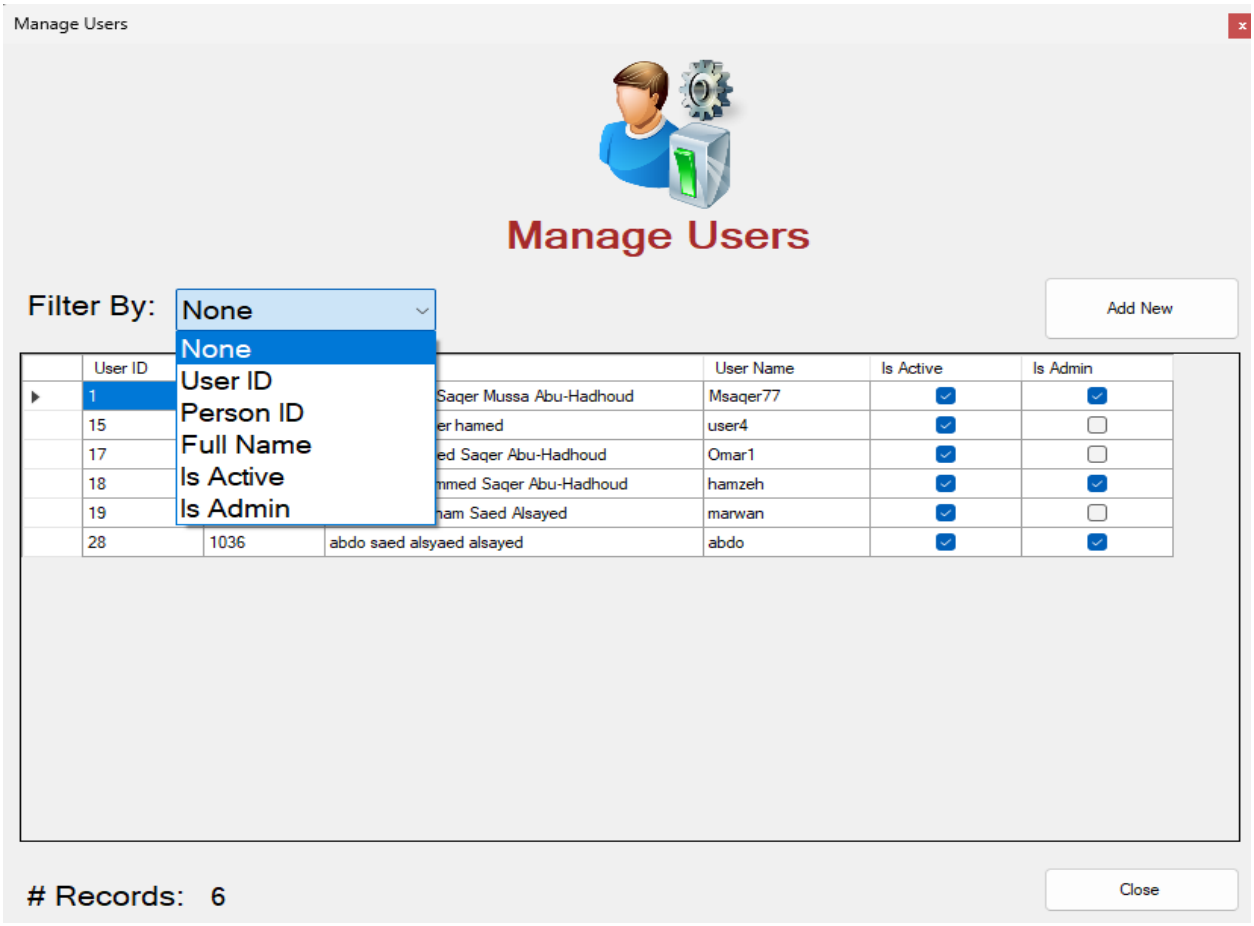
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Users Module**

Screen ID: **UI-USR-030**

Screenshot:



Screen Purpose:

- ✓ Display all system users
- ✓ Allow filtering and searching user records
- ✓ Provide administrative operations such as add, edit, delete, and change password

---

Functions:

- ✓ Load all users from the business layer
- ✓ Support numeric filtering for ID fields
- ✓ Support partial matching for Full Name
- ✓ Provide Yes / No / All filter for Is Active and Is Admin
- ✓ Open Show User Information screen
- ✓ Add new user (**Admin only**)
- ✓ Edit user information (**Admin only**)
- ✓ Delete user (**Admin only**)
- ✓ Change user password
- ✓ Send Email (Not implemented)
- ✓ Phone Call (Not implemented)

Validations: Validation Numeric Search on ID

Navigation: Main Menu → Users

Error Messages:

Error Code	Message	Trigger
ERR-099	Only Admin users can add new users.	Non-admin attempts Add
ERR-100	Only Admin users can edit user information.	Non-admin attempts Edit
ERR-101	Only Admin users can delete users.	Non-admin attempts Delete
ERR-102	Could not delete the user.	Delete operation failed
ERR-103	Failed to load users list.	Data retrieval failure
ERR-104	Only Admin users can add new users.	Non-admin attempts Add

---

Expected Outputs:

- ✓ Users displayed correctly in data grid
- ✓ Accurate record count shown
- ✓ Filtering works dynamically
- ✓ Admin operations execute successfully
- ✓ Confirmation messages displayed for successful actions
- ✓ List refreshed after modifications

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

31. Change User Password

Document Version: **1.0**

Last Updated: **14 Nov 2026**

Author: Abd Alrahman Saed

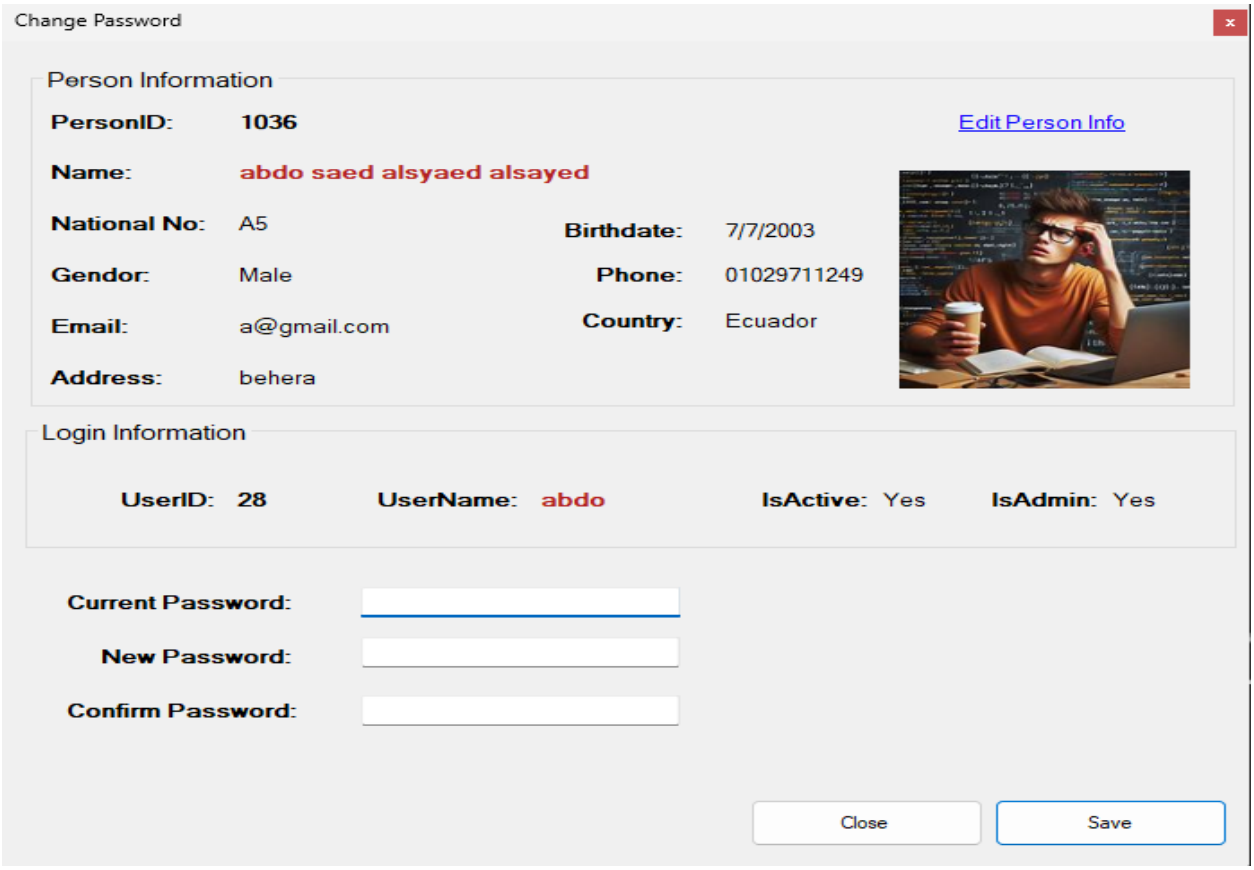
Reviewer: Dr. Heba Rashed

System Name: DVLD – Driving & Vehicle License Department System

Module Name: **Users Module**

Screen ID: **UI-USR-031**

Screenshot:



Screen Purpose:

Allow changing the password of an **existing system user**.

Functions:

- ✓ Load user information
- ✓ Display user details using User Card control
- ✓ Enter:
  - Current Password
  - New Password
  - Confirmed Password
- ✓ Save new password
- ✓ Display success or failure message

Validations:

- ✓ Current password is not empty
- ✓ Current password matches stored password
- ✓ New password is not empty
- ✓ Confirmed password matches new password

Navigation: Main Menu → Account Settings → Change Password

Error Messages:

Error Code	Message	Trigger
ERR-105	Please enter Current Password.	Empty current password
ERR-106	Current Password is incorrect.	Mismatch with stored password
ERR-107	Please enter New Password.	Empty new password
ERR-108	Please enter Confirmed Password.	Empty confirmed password

Error Code	Message	Trigger
ERR-109	Confirmed Password does not match New Password.	Confirmation mismatch
ERR-110	Some fields are not valid.	Validation failed
ERR-111	Could not change the password.	Change password operation failed

---

Expected Outputs:

- ✓ Password updated successfully
- ✓ Success confirmation message displayed

Changes Log:

Version	Date	Description of Change	Author
1.0	14 Nov 2026	Initial screen documentation	Abd Alrahman Saed

Conclusion:

The **DVLD – Driving & Vehicle License Department System** is a robust backend-driven solution developed to modernize and streamline the management of driving licenses, international licenses, applications, and user records within a centralized and secure platform.

The system was architected using a layered design approach that separates concerns between presentation, business logic, and data access layers.

By applying clean architecture principles, repository patterns, and structured database design, the solution ensures maintainability, scalability, and long-term sustainability.

Special attention was given to data integrity, validation mechanisms, secure authentication workflows, and optimized database operations to guarantee reliability and performance.

The modular structure of the system allows seamless feature expansion without impacting existing components.

From an engineering perspective, this project demonstrates strong backend development practices including structured entity modeling, controlled dependency management, secure authorization handling, and efficient database interaction.

In conclusion, the DVLD System represents a production-ready administrative platform that reflects modern backend engineering standards and showcases the practical implementation of software engineering concepts in a real-world governmental environment.

Developed & Engineered By:

Abd Alrahman Saed Alsayed

Computer Science Student